MTJA CO_2-TO-METHANOL PILOT

HAZOP REPORT

Hazard and operability study of the pilot trickle-bed reactor R-601
(teaching study on the R-601 operator training simulator)

Client	Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University
Project	MTJA CO_2 methanol pilot (ethanol-assisted low-temperature synthesis)
Unit	MTJA CO_2-to-methanol pilot · trickle-bed reactor R-601 (ethanol-assisted, 30 bar g, 150 °C)
Document no.	R601-SA-HZP-001
Title	HAZOP Report
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2; PFD 2407-PE-PFD-001
Revision	A

Rev	Date	Description	Prepared	Checked	Approved
A	2026-10-24	Issued for teaching	Apinan Soottitantawat		

This is a teaching study: the risk matrix is an example, the team names, signatures and session dates are left blank, and every alarm and trip setting is taken from the simulator (alarm list, trip list and constants) at the time of issue. It is not a substitute for the project HAZOP.

HAZOP close-out summary

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1	R01	N1.1	Add an absolute high-flow alarm on FT-201 (and FT-101) that works in every controller mode, and an H_2:CO_2 ratio deviation alarm on FY-100.	Instrument / Control		OPEN	Highest risk addressed: 6 Medium
N1 / N9	R02	N1.2 N1.4 N9.7	Raise PAL-203 and PAL-103 above the reactor pressure plus the MFC pressure drop (now 27 bar g, below the 30 bar g reactor), and review PALL-203/103 against the pressure at which the MFCs lose control.	Process / Instrument		OPEN	Highest risk addressed: 8 Medium
N1	R03	N1.3	Check FT-101 and FT-201 against a reference meter each campaign and reconcile the CO_2 feed with a GC carbon balance.	Operations / Instrument		OPEN	Highest risk addressed: 6 Medium
N1	R04	N1.4	Protect the CO_2 pack regulator against cooling and icing (heated or insulated PR-101) and fit a switch-over manifold.	Process / Mechanical		OPEN	Highest risk addressed: 8 Medium
N1	R05	N1.5	Confirm the pressure rating of the MFCs, XVs and tubing downstream of PR-202, PR-102 and PR-402 against full pack pressure, or add relief valves there (none in 2407-PE-RLB-001).	Process		OPEN	Highest risk addressed: 8 Medium
N1 / N2	R06	N1.6 N2.5	Put the check valves CK101, CK201, CK301, CK302, CK401 and CK601 in the inspection and proof-test plan.	Maintenance		OPEN	Highest risk addressed: 8 Medium
N1	R07	N1.7	Add a high alarm on FT-401 while the SIS is in RUN (N_2 passing into the feed).	Instrument		OPEN	Highest risk addressed: 3 Low
N1	R08	N1.8	Use gas-specific pack connections and labels, with a two-person check at every pack change.	Operations		OPEN	Highest risk addressed: 5 Medium
N1	R09	N1.9	Add the N_2 supply (PT-403 above PAL-403) to the start-up permissives, because every trip relies on the N_2 purge.	Instrument / Control		OPEN	Highest risk addressed: 9 Medium
N1	R10	N1.10	Check the supplier certificate (sulphur, chloride, moisture) for every CO_2 and H_2 pack received.	Operations		OPEN	Highest risk addressed: 4 Low
N2	R11	N2.1 N2.6	Confirm that the T-301 low-level switch stops P-301 (dry running), and add it to the C&E.	Instrument		OPEN	Highest risk addressed: 8 Medium
N2	R12	N2.2	Reconcile the ethanol used daily (T-301 level and the weigh test against the FT-301 totaliser) and add a high-output alarm on FIC-301.	Operations / Instrument		OPEN	Highest risk addressed: 6 Medium
N2	R13	N2.4	Re-set the P-301 discharge alarms and trips to the plant basis: the plant HMI runs P-301 on/off on PT-301 at 27–32 bar g, which is consistent with PSV-302 at 36 bar g (2407-PE-RLB-001), but the C&E (PALL-301 50, PAHH-301 62 bar g) and the OTS model (≈ 56 bar g behind a pressure-holding valve) follow an earlier design basis. Revise the C&E settings to the 27–32 bar g band and confirm the discharge rating; the OTS model is kept as validated.	Process / Instrument		OPEN	Highest risk addressed: 9 Medium
N2	R14	N2.5	Add gas break-through from the 30 bar g reactor through failed check valves to the T-301 / PSV-301 relief cases (only fire is listed).	Process		OPEN	Highest risk addressed: 8 Medium
N2	R15	N2.7	Write a T-301 filling procedure with the LS-301 high-level stop, and confirm the bund.	Operations		OPEN	Highest risk addressed: 9 Medium
N2 / N3 / N6 / N7	R16	N2.8 N3.4 N6.8 N7.6	Cover the fire cases of T-301, R-601/E-601, D-601 and T-601 in a fire and layout review (HAZID): spacing, extinguishers, drainage.	Safety		OPEN	Highest risk addressed: 8 Medium
N3	R17	N3.2	Confirm the TT-602 alarm settings: the C&E TAL 143 / TALL 135 °C sit above the 127 °C design inlet; the OTS uses a re-based 110 °C (assumed).	Process / Instrument		OPEN	Highest risk addressed: 3 Low

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N3	R18	N3.5	Document the existing gas detectors GD-001/GD-002 (type, gas, location, alarm set points; they read ppm on the plant HMI), confirm they cover E-601, R-601, D-601 and the gas packs, decide whether a detector should trip the feeds (no detector row on the C&E today), and leak-test with N_2 (then He) before every H_2 start-up.	Safety / Instrument		OPEN	Highest risk addressed: 10 High
N4	R19	N4.1 N4.2 N4.3 N4.4	Add high alarms on TT-603 and TT-604 and a 2oo3 high-high bed trip on TT-603/604/605 (today only TT-605 trips and TT-604 has no alarm).	Instrument / Process		OPEN	Highest risk addressed: 12 High
N4 / N9	R20	N4.1 N4.2 N9.6	Set the band-heater wall high alarm and trip at or below the 180 °C R-601 design temperature (now 260 °C, assumed) with an independent over-temperature cut-out per zone; review the E1 wall set point, which equals the design temperature.	Process / Electrical		OPEN	Highest risk addressed: 12 High
N4	R21	N4.2 N4.5	Fit a multipoint thermocouple in the bed so hot spots between TT-603, TT-604 and TT-605 are seen.	Instrument		OPEN	Highest risk addressed: 12 High
N4	R22	N4.3	Add a deviation alarm between TT-603/604/605 and allow TIC-604 cascade only while it is healthy.	Control		OPEN	Highest risk addressed: 9 Medium
N5	R23	N5.6	Add differential-pressure indication across F-601 (PT-601 − PT-608) with a high alarm.	Instrument		OPEN	Highest risk addressed: 6 Medium
N5	R24	N5.3	Add a chiller run-status alarm and a chilled-water supply temperature transmitter with a high alarm.	Instrument		OPEN	Highest risk addressed: 8 Medium
N5	R25	N5.5	Confirm the E-602 shell design pressure (10 bar g in 2407-ME-EQL-001, 4 bar g in 2407-PE-RLB-001) and the PSV-602 tube-rupture sizing; vent the chilled-water tank to a safe place.	Process / Mechanical		OPEN	Highest risk addressed: 8 Medium
N6	R26	N6.5	Carry out a vent dispersion study and hazardous-area classification for the cold vent (H_2-rich gas with CO), including depressure and PSV lifts.	Safety		OPEN	Highest risk addressed: 8 Medium
N6	R27	N6.3 N6.4	Give LIC-601 and the LALL-601 trip separate final elements (XV-601 is shared by the on/off level control and the trip).	Instrument		OPEN	Highest risk addressed: 9 Medium
N6	R28	N6.6	Calibrate the GC with a certified CO standard (not the CO_2 factor).	Instrument / Laboratory		OPEN	Highest risk addressed: 3 Low
N4 / N6	R29	N4.7 N6.7	Keep a positive N_2 blanket after shutdown and before unloading; review the vent for back-flow of air.	Process / Operations		OPEN	Highest risk addressed: 8 Medium
N2 / N4 / N5 / N6 / N7 / N9	R30	N2.1 N4.6 N5.1 N6.2 N6.4 N7.1 N7.3 N9.3	Review the start-up bypass: the low-low trips (PALL-301, PALL-601, LALL-601, LALL-602, FALL-701) are bypassed in READY; arm them automatically when their condition is healthy or after a time limit, and alarm when the SIS stays in READY with feed on.	Control / Safety		OPEN	Highest risk addressed: 9 Medium
N7	R31	N7.4	Add a T-601 high-high level that closes XV-602.	Instrument		OPEN	Highest risk addressed: 9 Medium
N7	R32	N7.5	Provide closed sampling for D-602/T-601 product and a methanol exposure procedure.	Operations / Safety		OPEN	Highest risk addressed: 9 Medium
N8	R33	N8.1	Set PALL-902 above the pressure at which the SOVs drop out (≈ 2 bar g in the model, assumed), so the air trip comes first and the first-out shows the air loss.	Instrument		OPEN	Highest risk addressed: 6 Medium
N8	R34	N8.3 N8.4	Confirm UPS supply for the SIS, DCS, MFCs and GC, and document the behaviour of every final element on loss of power.	Electrical / Instrument		OPEN	Highest risk addressed: 9 Medium
N9	R35	N9.1	Write a pressure-swing N_2 purge procedure with a portable O_2 check before H_2 is admitted.	Operations		OPEN	Highest risk addressed: 10 High
N9	R36	N9.2	Add a permissive: ethanol flowing (FT-301 above its low limit) before XV-201 can open in READY.	Control		OPEN	Highest risk addressed: 9 Medium

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 / N9	R37	N1.8 N9.8	Vent-purge the pigtail with N_2 at every pack changeover (written step).	Operations		OPEN	Highest risk addressed: 8 Medium
N4 / N9	R38	N4.7 N9.9	Write and HAZOP the catalyst loading, reduction and passivation/unloading procedures before first use.	Process / Operations		OPEN	Highest risk addressed: 8 Medium
N9	R39	N9.10	Use a permit to work with N_2 purge and a gas test before any equipment is opened.	Operations / Safety		OPEN	Highest risk addressed: 8 Medium
N9	R40	N9.11	Train operators on the simulator on the deviations of section 9 and on the start-up bypass, and record their response times.	Operations / Training		OPEN	Highest risk addressed: 6 Medium

HAZOP REPORT

MTJA CO_2-to-methanol pilot · trickle-bed reactor R-601 (ethanol-assisted, 30 bar g, 150 °C)
Document R601-SA-HZP-001 · Rev A

Approvals

Role	Name	Signature	Date
HAZOP leader (independent)			
Process engineer			
Operations representative			
Instrument / control engineer			
Safety engineer			
Project manager (approval)			

Names, signatures and dates are left blank on purpose: they are filled in by the real team. No person is named for a role in this teaching issue.

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Executive summary

The team examined 9 nodes of the R-601 pilot unit (gas feed, ethanol feed, E-601, R-601 with its band heaters, the condenser, the HP separator and vent, the LP side, the utilities and the procedures) with the eleven IEC 61882 guide words. 67 deviations were recorded and ranked on a 5 × 5 matrix: 12 Low, 51 Medium, 4 High and 0 Very high. 40 recommendations were raised. 10 of the deviations were run on the dynamic model with every alarm and trip active and no operator action.

Key findings

- N3.5 H_2 leak at a hot flange or fitting (10 High): H_2-rich gas at 30 bar g and ≈ 150 °C escapes; jet fire or explosion in the skid; flame almost invisible
- N4.1 Middle zone too hot (12 High): Hot spot in the middle of the bed far above the dew point: the bed dries, the ester route stops, Cu sinters (permanent loss of activity); tube metal above the 180 °C design temperature at 30 bar g with H_2: loss of strength, possible loss of containment and H_2 fire
- N4.2 Top zone too hot (12 High): Hot spot in the top zone below TT-603; bed dries and sinters; wall far above the design temperature; the bottom TC and TT-603 hardly move, so nothing alarms
- N9.1 H_2 admitted before the air is purged out (10 High): H_2/air flammable mixture in R-601 over hot catalyst: explosion or fast exotherm
- D1 More H_2 flow than set: No alarm; no trip in 29:00. FT-201 reached 58.2 NL/min, the vent flow FT-603 62.9 NL/min and TT-604 155.4 °C. The ±5 % flow alarms are relative to the set point and are suppressed outside AUTO, so a high H_2 flow in MAN is not alarmed at all (R01).
- D4 Middle zone too hot: First alarm TT-605 high at 15:53; TAHH-605 trips at 28:42. Before the trip TT-604 reached 213.1 °C with no alarm and the hottest bed cell 238.9 °C, far above the 180 °C R-601 design temperature; the only bed trip is on the bottom TC (R19, R20).
- D5 Top zone too hot: No alarm; no trip in 29:00. The hottest bed cell reached 189.0 °C and the wall 259.7 °C, while TT-603 moved only to 133.4 °C: the hot spot lies between the thermocouples and the wall alarm is set above the design temperature (R19, R20, R21).
- D10 No instrument air: First alarm FIC-101 low at 1:24; PAHH-301 trips at 1:28. The first-out is PAHH-301, not the air header: the valves drop out near 2 bar g, before the header reaches the 0.9 bar g PALL-902 setting (R33).

Actions by party

Discipline	Actions (a shared action counts for each party)
Instrument	19
Operations	12
Process	11
Safety	6
Control	5
Mechanical	2
Electrical	2
Maintenance	1
Laboratory	1
Training	1

1 Introduction

R-601 is the pilot trickle-bed reactor of the MTJA CO_2-to-methanol project. H_2 and CO_2 (5:1) with ethanol (EtOH:CO_2 = 4) pass over 17.34 kg of Cu/ZnO/Al_2O_3 at about 150 °C and 30 bar g; ethanol is both the solvent and a co-reactant (CO_2 + H_2 + EtOH → ethyl formate + H_2O; ethyl formate + 2H_2 → methanol + EtOH). The unit holds H_2 at 30 bar g next to electric heaters, and flammable and toxic liquids (ethanol, methanol); its SIS trips the feeds, stops the pump, purges with N_2 and switches off the heaters.
This report records a HAZOP of the unit after IEC 61882:2016. It was carried out as a teaching study on the R-601 operator training simulator: the safeguards quoted are the alarm and trip settings of the simulator (taken from C&E 2407-IN-CE-001 A2, with the assumptions listed in the simulator workbook), and selected deviations were checked dynamically on the model.

2 Scope

Inside the scope: the R-601 skid from the H_2, CO_2 and N_2 packs and the ethanol tank T-301 to the cold vent and the product tank T-601, with the instrument air and electrical supply, plus a procedure node for purge, start-up, shutdown, pack changeover and catalyst handling. Outside the scope: the gas supplier, the analyser house, the building and the T-602 waste route beyond the IBC. The catalyst loading, reduction and unloading procedures are not in the unit documents and are recorded as a gap.

Node	Name	Design intent	Rows
N1	H_2, CO_2 and N_2 gas feed	Deliver H_2 and CO_2 from the gas packs through the two-stage regulators and mass-flow controllers at the design flows (H_2:CO_2 = 5) to the E-601 inlet at ≈ 30 bar g; N_2 only for purge and pressurising.	10
N2	Ethanol feed	Pump ethanol from T-301 with the P-301 metering pump at the design rate (FIC-301 on FT-301) into the gas feed ahead of E-601; the pressure-holding valve keeps PT-301 ≈ 55.7 bar g.	9
N3	E-601 electric preheater	Heat the mixed gas and ethanol to ≈ 127.0 °C two-phase at the R-601 inlet with the element held at 150 °C by TIC-601.	6
N4	R-601 trickle bed and band heaters	Keep the 17.34 kg Cu/ZnO/Al_2O_3 bed at ≈ 130–151 °C, below the ≈ 152.5 °C dew point so the bed stays wetted, at ≈ 30 bar g, with the three band-heater zones E1/E2/E3 on wall control (TIC-611/612/613); TIC-604 cascade normally in MAN.	7
N5	F-601 filter, E-602 condenser and chilled water	Filter the reactor outlet (F-601) and condense the ethanol, methanol, water and esters in E-602 with chilled water at 15 °C, 67 L/min, so the condensate leaves at ≈ 30.4 °C to D-601.	6
N6	D-601 HP separator, BPV-601, vent and GC	Separate the condensate from the gas at 30 bar g; hold the pressure with BPV-601 and send ≈ 37.4 NL/min of gas to the cold vent past the GC; drain the liquid on/off to D-602 by LIC-601 through XV-601; XV-603 depressures to the cold vent.	8
N7	D-602 LP flash and T-601 product	Flash the D-601 liquid at ≈ 0.2 bar g in D-602 (BPV-602), drain it on/off by LIC-602 through XV-602 to the atmospheric product tank T-601; P-302 transfers product.	6
N8	Utilities: instrument air and electrical power	Supply instrument air at ≈ 4.0 bar g to the XVs and the pressure-holding valve, and power to the heaters, P-301, chiller, chilled-water pump, MFCs, GC and the SIS/DCS.	4
N9	Procedures: purge, start-up, shutdown, pack changeover, catalyst	Take the unit from cold, depressured and N_2-filled to the design point through the SIS states TRIPPED → READY → RUN; shut it down under N_2; change gas packs; load, reduce and unload the catalyst (procedures for the last are not in the unit documents).	11

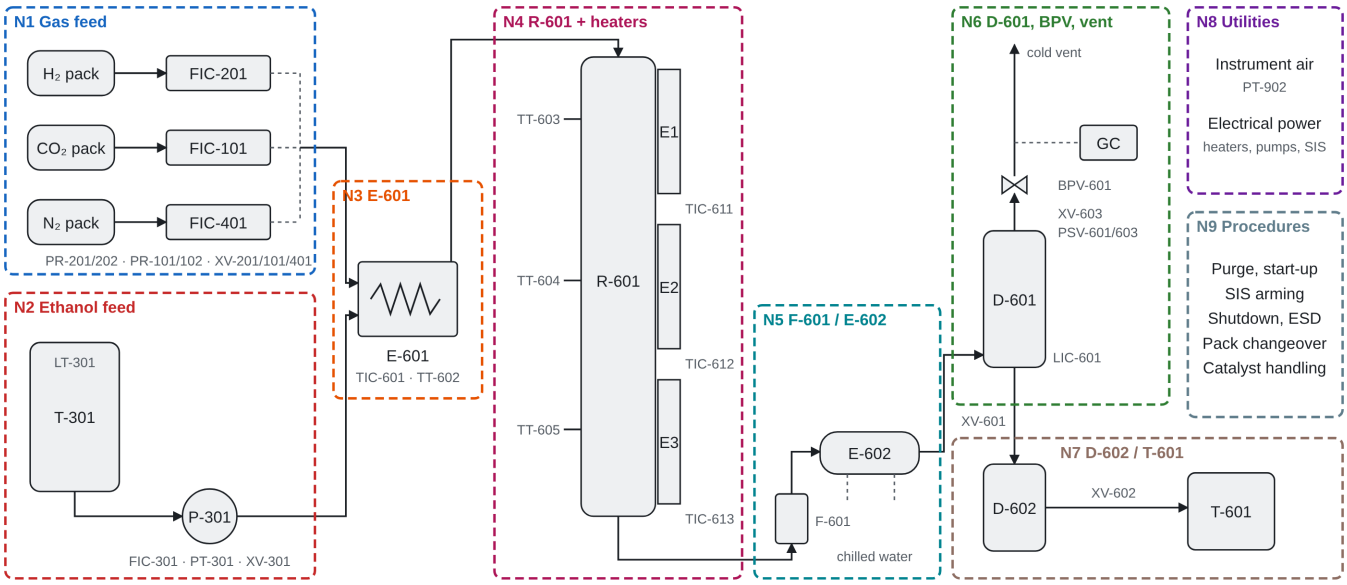


Figure 1. Node map of R-601 (simplified equipment, dashed node boundaries).

3 Methodology

IEC 61882 in four phases: definition (scope, objectives, team), preparation (documents, nodes, design intents, guide words, risk matrix), examination (node by node, parameter by parameter, guide word by guide word) and documentation and follow-up. Every deviation considered is written down (full recording). Consequences are judged with no credit for the safeguards; the likelihood is judged with the existing safeguards in place.

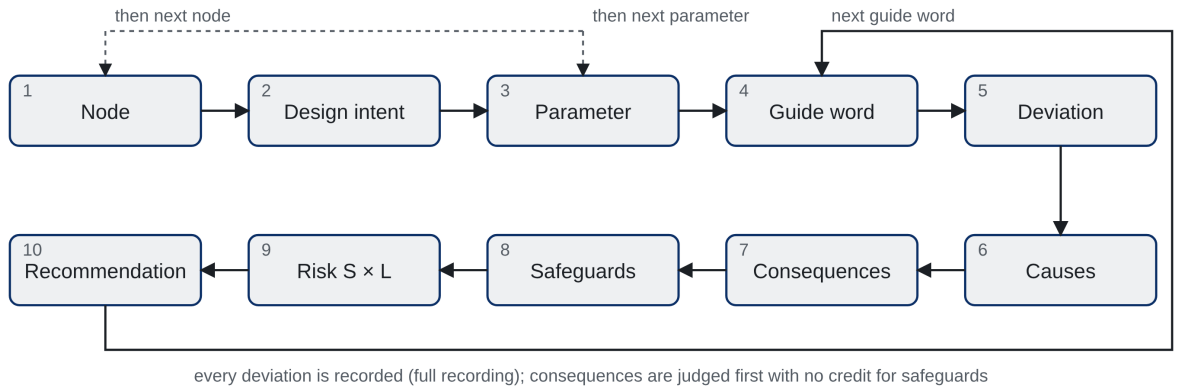


Figure 2. The examination loop.

3.1 Guide words and parameters

Guide word	Meaning	Example from this study
NO / NOT	Complete negation of the design intent	N2.1: Flow — No ethanol flow
MORE	Quantitative increase	N1.1: Flow — More H_2 flow than set
LESS	Quantitative decrease	N1.2: Flow — Less or no H_2 flow
AS WELL AS	Qualitative increase: something extra present	N1.7: Composition — N_2 in the feed during operation
PART OF	Qualitative decrease: only part of the intent achieved	N4.5: Reaction — Only part of the bed wetted
REVERSE	Logical opposite of the intent	N1.6: Flow — Back-flow from the reactor into a gas supply
OTHER THAN	Complete substitution, or something else happens	N1.8: Composition — Wrong gas connected to a feed line
EARLY	Relative to clock time: happens too soon	N9.2: Sequence — H_2 in before the ethanol flows
LATE	Relative to clock time: happens too late	N9.3: Time — SIS left in READY
BEFORE	Relative to order or sequence: comes before	N9.1: Sequence — H_2 admitted before the air is purged out
AFTER	Relative to order or sequence: comes after	N9.6: Sequence — Heaters left on after the feeds stop

Parameter	Units / meaning
Flow	NL/min (gas), mL/min (ethanol), L/min (chilled water)
Pressure	bar g
Temperature	°C (element, bed, wall, condensate)
Level	% (T-301, D-601, D-602, T-601)
Composition	mol % (feed ratio, vent gas, product)
Reaction	activity, wetting, route
Air / power	instrument-air header, electrical supply
Sequence	order of steps in a procedure
Time	how soon or late a step happens
Maintenance	isolation, opening, catalyst handling

3.2 Risk matrix

S \ L	1 Remote	2 Unlikely	3 Possible	4 Likely	5 Frequent
5 Catastrophic	**1** (5)	**2** (10)	· (15)	· (20)	· (25)
4 Major	**1** (4)	**12** (8)	**2** (12)	· (16)	· (20)
3 Moderate	· (3)	**5** (6)	**15** (9)	· (12)	· (15)
2 Minor	· (2)	**4** (4)	**14** (6)	**4** (8)	· (10)
1 Negligible	· (1)	**1** (2)	**4** (3)	**2** (4)	· (5)

Cell: number of worksheet rows and (R = S × L). Rows: 67; Low 12, Medium 51, High 4, Very high 0. The same bands are used by the HAZID.

S	Level	People	Asset / production	Environment
1	Negligible	No injury	Production loss < 1 day, no damage	No release
2	Minor	First aid	Off-spec product, minor repair	Small release inside the bund or vent
3	Moderate	Medical treatment	Repair of one item, days lost	Reportable release on site
4	Major	Lost-time or serious injury	Major damage to R-601 or the skid	Release off site, short term
5	Catastrophic	Fatality	Fire or explosion destroying the unit	Major off-site impact

L	Level	Description	Frequency
1	Remote	Not known in the industry	$< 10^{-4}$ /yr
2	Unlikely	Known in the industry	10^{-4} – 10^{-3} /yr
3	Possible	Has happened in the company / similar pilots	10^{-3} – 10^{-2} /yr
4	Likely	Expected once in the life of the unit	10^{-2} – 10^{-1} /yr
5	Frequent	Several times per year	$> 10^{-1}$ /yr

Risk	R	Action
Low	1–4	Acceptable; manage by procedure and review at revalidation
Medium	5–9	Tolerable if ALARP; action within the campaign, owner named
High	10–16	Not acceptable as it stands; action before the next start-up, LOPA
Very high	20–25	Stop: do not operate until the risk is reduced

4 Sessions

4.1 Schedule

Session	Nodes	Date	Attendees
1	N1, N2, N3		
2	N4, N5		
3	N6, N7, N8		
4	N9 and review of recommendations		

4.2 Team

Role	Contribution	Name
HAZOP leader (independent)	Guides the team through nodes, parameters and guide words; keeps to the method; not from the design team	
Scribe	Records every deviation (full recording) in the worksheet	
Process engineer	Design intent, the heat and mass balance, the kinetics and the operating window	
Operations representative	How the unit is really run: procedures, pack changes, alarms the operators see	
Instrument / control engineer	Alarms, trips, fail positions, the C&E and the SIS	
Safety engineer	Risk matrix, relief and fire cases, follow-up to LOPA	
Specialists as needed	Catalyst supplier, electrical (band heaters, power), mechanical (pressure ratings)	

4.3 Documents reviewed

Document	Where in the OTS	Used for
P&ID 2407-PE-PID-001 A6	P&ID tab	Node boundaries, tags, valves, relief
PFD 2407-PE-PFD-001 and HMB 2407-PE-HMB-001	Process tab	Design intent, flows, conditions
C&E 2407-IN-CE-001 A2	C&E matrix tab	Alarm and trip settings, effects
Instrument list 2407-IN-INL-001 R4	P&ID / C&E tabs	Instrument tags and ranges
Relief-load 2407-PE-RLB-001 A3	this tab (safeguards)	PSV set pressures and governing cases
Equipment list 2407-ME-EQL-001 A1	Design tab	Design pressures and temperatures
Kinetic report (pilot TBR scale-up)	Operating windows tab	Dew point, fitted window, hot spots
SIS logic, permissives, trips	SIS tab	Arming, bypass, first-out
Operating procedures	Exercises tab (procedures)	Start-up, shutdown, ESD, pack changeover
Dynamic model	this tab, part 9	Deviation runs with the safeguards active
SRS of SIF (when issued)	SRS of SIF tab	Trip requirements for rows that rely on a SIF

4.4 Guide-word coverage

Node	NO	MORE	LESS	AS WELL	PART OF	REVERSE	OTHER	EARLY	LATE	BEFORE	AFTER	Rows
N1	·	2	4	1	·	1	2	·	·	·	·	10
N2	1	4	2	·	·	1	1	·	·	·	·	9
N3	1	2	1	1	·	·	1	·	·	·	·	6
N4	·	3	2	·	1	·	1	·	·	·	·	7
N5	1	2	3	·	·	·	·	·	·	·	·	6

Node	NO	MORE	LESS	AS WELL	PART OF	REVERSE	OTHER	EARLY	LATE	BEFORE	AFTER	Rows
N6	·	3	2	1	·	1	1	·	·	·	·	8
N7	·	4	1	·	·	·	1	·	·	·	·	6
N8	2	1	1	·	·	·	·	·	·	·	·	4
N9	·	·	·	·	·	·	2	2	3	3	1	11
All	5	21	16	3	1	3	9	2	3	3	1	67

5 Findings

5.1 Risk profile per node

Node	Name	Rows	Low	Medium	High	Very high	Max R
N1	H_2, CO_2 and N_2 gas feed	10	2	8	0	0	9
N2	Ethanol feed	9	1	8	0	0	9
N3	E-601 electric preheater	6	2	3	1	0	10
N4	R-601 trickle bed and band heaters	7	1	4	2	0	12
N5	F-601 filter, E-602 condenser and chilled water	6	1	5	0	0	9
N6	D-601 HP separator, BPV-601, vent and GC	8	2	6	0	0	9
N7	D-602 LP flash and T-601 product	6	0	6	0	0	9
N8	Utilities: instrument air and electrical power	4	1	3	0	0	9
N9	Procedures: purge, start-up, shutdown, pack changeover, catalyst	11	2	8	1	0	10

5.2 High-ranked rows

Ref	Deviation	Causes	Consequences	Risk	Rec.
N3.5	H_2 leak at a hot flange or fitting	Thermal cycling of the E-601 and R-601 connections; gasket failure; fitting not tightened after maintenance	H_2-rich gas at 30 bar g and ≈ 150 °C escapes; jet fire or explosion in the skid; flame almost invisible	**10 High**	R18
N4.1	Middle zone too hot	E2 band-heater SCR failing on at full power; TIC-612 output failing high	Hot spot in the middle of the bed far above the dew point: the bed dries, the ester route stops, Cu sinters (permanent loss of activity); tube metal above the 180 °C design temperature at 30 bar g with H_2: loss of strength, possible loss of containment and H_2 fire	**12 High**	R19 R20
N4.2	Top zone too hot	E1 band-heater SCR or TIC-611 output failing high; TIC-611 in MAN at a high output	Hot spot in the top zone below TT-603; bed dries and sinters; wall far above the design temperature; the bottom TC and TT-603 hardly move, so nothing alarms	**12 High**	R19 R20 R21
N9.1	H_2 admitted before the air is purged out	Start-up after maintenance with the unit only pressurised with N_2, not purged to a low O_2 content; step skipped	H_2/air flammable mixture in R-601 over hot catalyst: explosion or fast exotherm	**10 High**	R35

5.3 Recommendation register

ID	Recommendation	Node	Rows	Highest risk	Action by
R01	Add an absolute high-flow alarm on FT-201 (and FT-101) that works in every controller mode, and an H_2:CO_2 ratio deviation alarm on FY-100.	N1	N1.1	**6 Medium**	Instrument / Control
R02	Raise PAL-203 and PAL-103 above the reactor pressure plus the MFC pressure drop (now 27 bar g, below the 30 bar g reactor), and review PALL-203/103 against the pressure at which the MFCs lose control.	N1, N9	N1.2 N1.4 N9.7	**8 Medium**	Process / Instrument
R03	Check FT-101 and FT-201 against a reference meter each campaign and reconcile the CO_2 feed with a GC carbon balance.	N1	N1.3	**6 Medium**	Operations / Instrument
R04	Protect the CO_2 pack regulator against cooling and icing (heated or insulated PR-101) and fit a switch-over manifold.	N1	N1.4	**8 Medium**	Process / Mechanical
R05	Confirm the pressure rating of the MFCs, XVs and tubing downstream of PR-202, PR-102 and PR-402 against full pack pressure, or add relief valves there (none in 2407-PE-RLB-001).	N1	N1.5	**8 Medium**	Process
R06	Put the check valves CK101, CK201, CK301, CK302, CK401 and CK601 in the inspection and proof-test plan.	N1, N2	N1.6 N2.5	**8 Medium**	Maintenance
R07	Add a high alarm on FT-401 while the SIS is in RUN (N_2 passing into the feed).	N1	N1.7	**3 Low**	Instrument

ID	Recommendation	Node	Rows	Highest risk	Action by
R08	Use gas-specific pack connections and labels, with a two-person check at every pack change.	N1	N1.8	**5 Medium**	Operations
R09	Add the N_2 supply (PT-403 above PAL-403) to the start-up permissives, because every trip relies on the N_2 purge.	N1	N1.9	**9 Medium**	Instrument / Control
R10	Check the supplier certificate (sulphur, chloride, moisture) for every CO_2 and H_2 pack received.	N1	N1.10	**4 Low**	Operations
R11	Confirm that the T-301 low-level switch stops P-301 (dry running), and add it to the C&E.	N2	N2.1 N2.6	**8 Medium**	Instrument
R12	Reconcile the ethanol used daily (T-301 level and the weigh test against the FT-301 totaliser) and add a high-output alarm on FIC-301.	N2	N2.2	**6 Medium**	Operations / Instrument
R13	Re-set the P-301 discharge alarms and trips to the plant basis: the plant HMI runs P-301 on/off on PT-301 at 27–32 bar g, which is consistent with PSV-302 at 36 bar g (2407-PE-RLB-001), but the C&E (PALL-301 50, PAHH-301 62 bar g) and the OTS model (≈ 56 bar g behind a pressure-holding valve) follow an earlier design basis. Revise the C&E settings to the 27–32 bar g band and confirm the discharge rating; the OTS model is kept as validated.	N2	N2.4	**9 Medium**	Process / Instrument
R14	Add gas break-through from the 30 bar g reactor through failed check valves to the T-301 / PSV-301 relief cases (only fire is listed).	N2	N2.5	**8 Medium**	Process
R15	Write a T-301 filling procedure with the LS-301 high-level stop, and confirm the bund.	N2	N2.7	**9 Medium**	Operations
R16	Cover the fire cases of T-301, R-601/E-601, D-601 and T-601 in a fire and layout review (HAZID): spacing, extinguishers, drainage.	N2, N3, N6, N7	N2.8 N3.4 N6.8 N7.6	**8 Medium**	Safety
R17	Confirm the TT-602 alarm settings: the C&E TAL 143 / TALL 135 °C sit above the 127 °C design inlet; the OTS uses a re-based 110 °C (assumed).	N3	N3.2	**3 Low**	Process / Instrument
R18	Document the existing gas detectors GD-001/GD-002 (type, gas, location, alarm set points; they read ppm on the plant HMI), confirm they cover E-601, R-601, D-601 and the gas packs, decide whether a detector should trip the feeds (no detector row on the C&E today), and leak-test with N_2 (then He) before every H_2 start-up.	N3	N3.5	**10 High**	Safety / Instrument
R19	Add high alarms on TT-603 and TT-604 and a 2oo3 high-high bed trip on TT-603/604/605 (today only TT-605 trips and TT-604 has no alarm).	N4	N4.1 N4.2 N4.3 N4.4	**12 High**	Instrument / Process
R20	Set the band-heater wall high alarm and trip at or below the 180 °C R-601 design temperature (now 260 °C, assumed) with an independent over-temperature cut-out per zone; review the E1 wall set point, which equals the design temperature.	N4, N9	N4.1 N4.2 N9.6	**12 High**	Process / Electrical
R21	Fit a multipoint thermocouple in the bed so hot spots between TT-603, TT-604 and TT-605 are seen.	N4	N4.2 N4.5	**12 High**	Instrument
R22	Add a deviation alarm between TT-603/604/605 and allow TIC-604 cascade only while it is healthy.	N4	N4.3	**9 Medium**	Control
R23	Add differential-pressure indication across F-601 (PT-601 − PT-608) with a high alarm.	N5	N5.6	**6 Medium**	Instrument
R24	Add a chiller run-status alarm and a chilled-water supply temperature transmitter with a high alarm.	N5	N5.3	**8 Medium**	Instrument
R25	Confirm the E-602 shell design pressure (10 bar g in 2407-ME-EQL-001, 4 bar g in 2407-PE-RLB-001) and the PSV-602 tube-rupture sizing; vent the chilled-water tank to a safe place.	N5	N5.5	**8 Medium**	Process / Mechanical
R26	Carry out a vent dispersion study and hazardous-area classification for the cold vent (H_2-rich gas with CO), including depressure and PSV lifts.	N6	N6.5	**8 Medium**	Safety
R27	Give LIC-601 and the LALL-601 trip separate final elements (XV-601 is shared by the on/off level control and the trip).	N6	N6.3 N6.4	**9 Medium**	Instrument
R28	Calibrate the GC with a certified CO standard (not the CO_2 factor).	N6	N6.6	**3 Low**	Instrument / Laboratory
R29	Keep a positive N_2 blanket after shutdown and before unloading; review the vent for back-flow of air.	N4, N6	N4.7 N6.7	**8 Medium**	Process / Operations
R30	Review the start-up bypass: the low-low trips (PALL-301, PALL-601, LALL-601, LALL-602, FALL-701) are bypassed in READY; arm them automatically when their condition is healthy or after a time limit, and alarm when the SIS stays in READY with feed on.	N2, N4, N5, N6, N7, N9	N2.1 N4.6 N5.1 N6.2 N6.4 N7.1 N7.3 N9.3	**9 Medium**	Control / Safety
R31	Add a T-601 high-high level that closes XV-602.	N7	N7.4	**9 Medium**	Instrument

ID	Recommendation	Node	Rows	Highest risk	Action by
R32	Provide closed sampling for D-602/T-601 product and a methanol exposure procedure.	N7	N7.5	**9 Medium**	Operations / Safety
R33	Set PALL-902 above the pressure at which the SOVs drop out (≈ 2 bar g in the model, assumed), so the air trip comes first and the first-out shows the air loss.	N8	N8.1	**6 Medium**	Instrument
R34	Confirm UPS supply for the SIS, DCS, MFCs and GC, and document the behaviour of every final element on loss of power.	N8	N8.3 N8.4	**9 Medium**	Electrical / Instrument
R35	Write a pressure-swing N_2 purge procedure with a portable O_2 check before H_2 is admitted.	N9	N9.1	**10 High**	Operations
R36	Add a permissive: ethanol flowing (FT-301 above its low limit) before XV-201 can open in READY.	N9	N9.2	**9 Medium**	Control
R37	Vent-purge the pigtail with N_2 at every pack changeover (written step).	N1, N9	N1.8 N9.8	**8 Medium**	Operations
R38	Write and HAZOP the catalyst loading, reduction and passivation/unloading procedures before first use.	N4, N9	N4.7 N9.9	**8 Medium**	Process / Operations
R39	Use a permit to work with N_2 purge and a gas test before any equipment is opened.	N9	N9.10	**8 Medium**	Operations / Safety
R40	Train operators on the simulator on the deviations of section 9 and on the start-up bypass, and record their response times.	N9	N9.11	**6 Medium**	Operations / Training

5.4 Dynamic check on the simulator

10 deviations were imposed on copies of the design state 60 s into a run and followed for up to 30 min (stopping 60 s after a trip) with all alarms and trips active and no operator action (model runs of 2026-10-23). Times are mm:ss after the deviation started.

ID	Row	Deviation and how it was imposed	First alarms	Trip first-out	Key tags in the run	Hottest bed cell
D1	N1.1	More H_2 flow than set — FIC-201 put in MAN at 60 % output (MFC or controller failing high)	none	no trip in 29:00	FT-201 32.7 to 58.2 NL/min (design 32.7) FT-603 37.4 to 62.9 NL/min (design 37.4) TT-604 148.3 to 155.4 °C (design 148.3)	160.0 °C (design 151.6)
D2	N2.1	No ethanol flow — P-301 metering pump stops (motor trip)	FIC-301 low at 0:01	PALL-301 at 0:08	PT-301 0.2 to 55.7 bar g (design 55.7) FT-301 0.0 to 68.2 mL/min (design 68.2) TT-605 150.9 to 151.4 °C (design 151.4)	152.2 °C (design 151.6)
D3	N3.1	E-601 element and outlet too hot — TIC-601 put in MAN at 100 % output (SCR or controller failing high)	TIC-601 high at 0:06	TAHH-601 at 0:15	TT-601 150.0 to 169.7 °C (design 150.0) TT-602 127.0 to 150.2 °C (design 127.0) PT-601 19.93 to 30.18 bar g (design 30.18)	151.9 °C (design 151.6)
D4	N4.1	Middle zone too hot — E2 band-heater SCR fails on at full power	TT-605 high at 15:53	TAHH-605 at 28:42	TT-604 148.3 to 213.1 °C (design 148.3) bed max (model) 151.6 to 238.9 °C (design 151.6) TT-605 151.4 to 167.0 °C (design 151.4)	238.9 °C (design 151.6)
D5	N4.2	Top zone too hot — TIC-611 put in MAN at 100 % output (E1 SCR or controller failing high)	none	no trip in 29:00	TT-603 130.2 to 133.4 °C (design 130.2) bed max (model) 151.6 to 189.0 °C (design 151.6) TIC-611 wall 180.0 to 259.7 °C (design 180.0)	189.0 °C (design 151.6)
D6	N5.1	No chilled-water flow — Chilled-water pump stops	FT-701 low at 0:01 TT-609 high at 0:05	FALL-701 at 0:06	FT-701 0.0 to 67.0 L/min (design 67.0) TT-609 30.4 to 116.2 °C (design 30.4) PT-601 20.02 to 30.18 bar g (design 30.18)	151.9 °C (design 151.6)
D7	N5.3	Chilled water too warm — Chiller compressor stops; the pump keeps running	TT-609 high at 16:18	no trip in 29:00	TT-609 30.4 to 33.2 °C (design 30.4) FT-603 37.4 to 37.4 NL/min (design 37.4) AY-620 MeOH yield 10.11 to 10.11 % (design 10.11)	151.6 °C (design 151.6)

ID	Row	Deviation and how it was imposed	First alarms	Trip first-out	Key tags in the run	Hottest bed cell
D8	N6.1	D-601 and R-601 pressure high — BPV-601 sticking closed (capacity to zero over 60 s)	PT-601 high at 2:32 PT-608 high at 2:37	PAHH-601 at 3:34	PT-601 21.67 to 32.96 bar g (design 30.18) PT-608 21.11 to 32.87 bar g (design 30.09) FT-603 0.0 to 723.8 NL/min (design 37.4)	152.4 °C (design 151.6)
D9	N7.1	Gas blow-by into D-602 — XV-601 held open (LIC-601 in MAN at 100 %) with the SIS in READY, low-low trips bypassed as at start-up	LIC-602 high at 2:41 LIC-601 low at 2:56	LAHH-602 at 5:02	LT-601 0.5 to 47.8 % (design 47.7) LT-602 35.0 to 80.2 % (design 35.0) D-602 pressure (model) 0.20 to 0.26 bar g (design 0.20)	152.0 °C (design 151.6)
D10	N8.1	No instrument air — Instrument-air compressor stops	FIC-101 low at 1:24 FIC-201 low at 1:24 FIC-301 low at 1:24	PAHH-301 at 1:28	PT-902 1.19 to 4.00 bar g (design 4.00) PT-301 6.7 to 61.7 bar g (design 55.7) PT-601 19.55 to 30.18 bar g (design 30.18)	152.0 °C (design 151.6)

What the runs tell the team

- D1 More H_2 flow than set: No alarm; no trip in 29:00. FT-201 reached 58.2 NL/min, the vent flow FT-603 62.9 NL/min and TT-604 155.4 °C. The ±5 % flow alarms are relative to the set point and are suppressed outside AUTO, so a high H_2 flow in MAN is not alarmed at all (R01).
- D2 No ethanol flow: First alarm FIC-301 low at 0:01; PALL-301 trips at 0:08. TT-605 stayed at or below 151.4 °C: the low-low pressure trip acts before the bed dries (but only in RUN, R30).
- D3 E-601 element and outlet too hot: First alarm TIC-601 high at 0:06; TAHH-601 trips at 0:15. TT-602 reached 150.2 °C before the trip; the element trip acts well before the bed sees a hotter feed.
- D4 Middle zone too hot: First alarm TT-605 high at 15:53; TAHH-605 trips at 28:42. Before the trip TT-604 reached 213.1 °C with no alarm and the hottest bed cell 238.9 °C, far above the 180 °C R-601 design temperature; the only bed trip is on the bottom TC (R19, R20).
- D5 Top zone too hot: No alarm; no trip in 29:00. The hottest bed cell reached 189.0 °C and the wall 259.7 °C, while TT-603 moved only to 133.4 °C: the hot spot lies between the thermocouples and the wall alarm is set above the design temperature (R19, R20, R21).
- D6 No chilled-water flow: First alarm FT-701 low at 0:01; FALL-701 trips at 0:06. TT-609 reached 116.2 °C as the condenser lost its duty; the trip is fast, but FALL-701 is armed in RUN only (R30).
- D7 Chilled water too warm: First alarm TT-609 high at 16:18; no trip in 29:00. TT-609 rose only to 33.2 °C, below the 34 °C trip: the unit runs on with warm condensate and more solvent to the vent, and nothing tells the operator the chiller has stopped (R24).
- D8 D-601 and R-601 pressure high: First alarm PT-601 high at 2:32; PAHH-601 trips at 3:34. The inlet pressure peaked at 32.96 bar g (5 s samples), below the 36 bar g PSV-601 set pressure; XV-603 opened with the trip.
- D9 Gas blow-by into D-602: First alarm LIC-602 high at 2:41; LAHH-602 trips at 5:02. D-601 drained to 0.5 % before D-602 reached its high-high level; the model D-602 pressure stayed at 0.26 bar g, below the 2 bar g PSV-606. In READY the LALL-601 trip is bypassed and the protection is the D-602 high-high level (R30).
- D10 No instrument air: First alarm FIC-101 low at 1:24; PAHH-301 trips at 1:28. The first-out is PAHH-301, not the air header: the valves drop out near 2 bar g, before the header reaches the 0.9 bar g PALL-902 setting (R33).

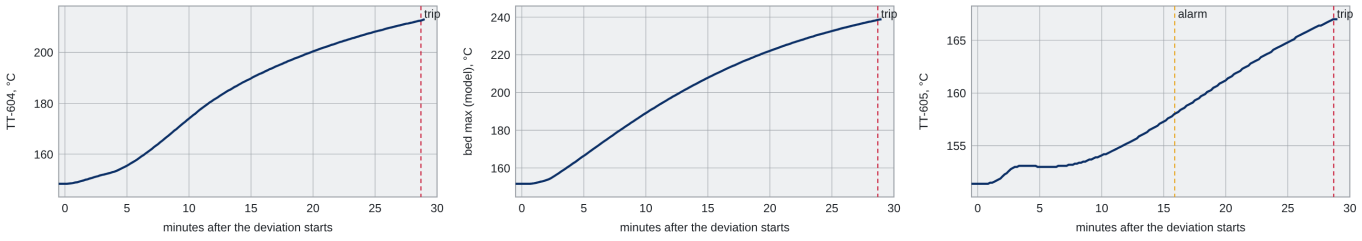


Figure 3. D4 Middle zone too hot: E2 band-heater SCR fails on at full power (dashed lines: first alarm of the tag, trip).

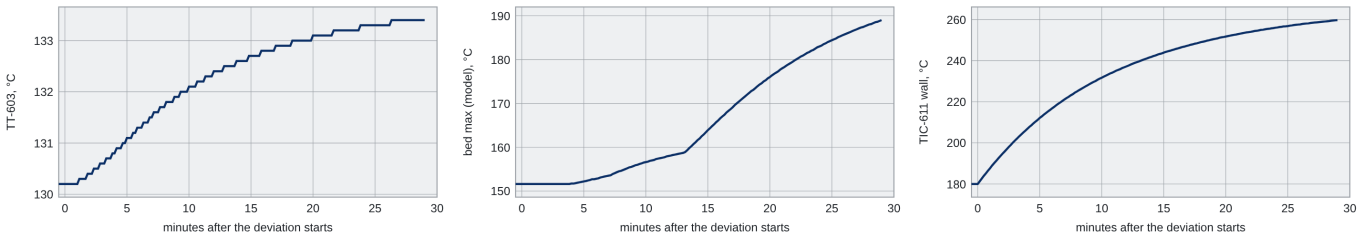


Figure 4. D5 Top zone too hot: TIC-611 put in MAN at 100 % output (E1 SCR or controller failing high) (dashed lines: first alarm of the tag, trip).

5.5 Follow-up

- Each recommendation has an owner (the discipline in the register), a target date and a written response; the HAZOP leader closes it when the response is accepted (Appendix 3).
- The 4 High rows (N3.5, N4.1, N4.2, N9.1) and every row whose main safeguard is a trip go to LOPA to decide whether the trip needs a SIL rating.
- Changes to set points, trips, bypasses, the catalyst or the procedures go through management of change against these worksheets.
- Revalidate at least every 5 years (assumed), after an incident and before a scale-up.
- Train the operators on the simulator on the deviations of 5.4.

Appendix 1 Worksheets

N1 H_2, CO_2 and N_2 gas feed

Project	MTJA CO_2 methanol pilot · R-601	Node	N1 H_2, CO_2 and N_2 gas feed
Design intent	Deliver H_2 and CO_2 from the gas packs through the two-stage regulators and mass-flow controllers at the design flows (H_2:CO_2 = 5) to the E-601 inlet at ≈ 30 bar g; N_2 only for purge and pressurising.	Drawing	2407-PE-PID-001 A6 (feed sheet); 2407-IN-CE-001 A2 rows 002–006, 011–014, 018
Conditions	Supply 35 bar g after the regulators; H_2 32.70 NL/min, CO_2 6.54 NL/min; ambient temperature	Instruments	PT-203, PT-103, PT-403; FIC-201 (FT/FV-201), FIC-101 (FT/FV-101), FIC-401 (FT/FV-401); ratio station FY-100
Equipment	H_2, CO_2 and N_2 packs; PR-201/202, PR-101/102, PR-401/402; filters F201/F101/F401; check valves CK201/CK101/CK401; XV-201, XV-101, XV-401	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	MORE	More H_2 flow than set	• FIC-201 MFC or its output failing high • controller left in MAN at a high output • set-point entry error • FY-100 ratio set wrong	H_2:CO_2 well above 5; more gas through BPV-601 to the cold vent (larger H_2-rich vent plume); bed temperature rises a few degrees; operation outside the fitted kinetic window	• FAH-201 at 105 % of SP (only while FIC-201 is in AUTO) • PAH-601 32 bar g • TAH-605 158 °C • MFC full scale 100 NL/min limits the flow • BPV-601 holds the pressure	2	3	**6 Medium**	R01	Instrument
N1.2	Flow	LESS	Less or no H_2 flow	• H_2 pack running down • PR-202 failing closed • XV-201 closing (solenoid or air fault) • MFC failing low • filter F201 plugging	H_2:CO_2 falls, methanol yield falls while the ester route continues; MFC loses its pressure drop before any alarm because PAL-203 is below the 30 bar g reactor pressure; if the supply falls below the reactor, back-flow (see N1.6)	• FAL-201 at 95 % of SP • PAL-203 27 bar g • PALL-203 trip at 24 bar g (2 s delay) • spare H_2 pack (PACK B) with changeover in the field	2	4	**8 Medium**	R02	Process
N1.3	Flow	LESS	Less real CO_2 than indicated	• FT-101 calibration drifting high, so FIC-101 cuts the real CO_2 • MFC failing low • PR-102 freezing	H_2:CO_2 above 5 unseen by the operator; yield per CO_2 and the reported conversion are wrong; the FAL/FAH-101 deviation alarms use the same transmitter and cannot see the error	• FAL/FAH-101 ±5 % of SP (same transmitter) • GC AI-611 CO_2 in the vent gas every 240 s • AY-620 yield soft sensor • PAL-103 27 bar g	2	3	**6 Medium**	R03	Operations
N1.4	Pressure	LESS	Low CO_2 supply pressure	• CO_2 pack cold (cylinder cooling during draw-off), pack empty, regulator PR-101 icing	CO_2 flow falls, H_2:CO_2 rises; regulator icing can make the pressure swing; loss of CO_2 trips the unit	• PAL-103 27 bar g • PALL-103 trip at 24 bar g (2 s delay, opens XV-603) • FAL-101 at 95 % of SP	2	4	**8 Medium**	R02 R04	Process
N1.5	Pressure	MORE	Full pack pressure downstream of the regulators	• First and second-stage regulator (PR-202, PR-102 or PR-402) failing open or set wrong	MFCs, XVs and tubing see full pack pressure; R-601 pressure rises if the MFC passes; overpressure of components rated for the regulated pressure; H_2 release at a fitting	• Two regulators in series • PAH-601 32 bar g • PAHH-601 trip at 33 bar g (1 s delay, opens XV-603) • PSV-601 36 bar g on E-601/R-601 (no relief valve on the feed manifold in 2407-PE-RLB-001)	4	2	**8 Medium**	R05	Process
N1.6	Flow	REVERSE	Back-flow from the reactor into a gas supply	• Supply pressure below the reactor (empty pack with XV open • CO_2 supply drawn down) • MFC open during a pressure test	Reactor gas and ethanol vapour into the CO_2 or N_2 manifold and packs; flammable gas in the CO_2 system; contamination of the pack	• Check valves CK201, CK101, CK401 • PALL-203 trip at 24 bar g (2 s delay) • PALL-103 trip at 24 bar g (2 s delay, opens XV-603) • the trips close XV-201/101, which also fail closed	3	2	**6 Medium**	R06	Maintenance

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N1.7	Composition	AS WELL AS	N_2 in the feed during operation	• XV-401 or FV-401 passing • FIC-401 left with a set point after start-up	N_2 dilutes the feed, H_2 partial pressure and yield fall slowly; vent rate rises; no safety consequence	• FT-401 indication • GC AI-613 N_2 in the vent gas • start-up step "stop the N_2"	1	3	**3 Low**	R07	Instrument
N1.8	Composition	OTHER THAN	Wrong gas connected to a feed line	• Wrong pack or cylinder connected at changeover (air, O_2 or another gas on the H_2 or CO_2 line) • pigtail not purged	Air or O_2 with H_2 over hot reduced Cu catalyst: flammable mixture, fast exotherm, catalyst oxidation, possible explosion in R-601	• Gas-specific pack connections (assumed) • pack labels • GC on the vent gas (after the event)	5	1	**5 Medium**	R08 R37	Operations
N1.9	Pressure	LESS	Low N_2 supply pressure	• N_2 pack running down unnoticed • PR-402 failing closed	The N_2 purge that every trip relies on is not available: H_2 stays in R-601 after a trip; no inert for a shutdown	• PAL-403 24 bar g • XV-401 and FV-401 fail open (purge path open on a trip). N_2 is not among the start-up permissives	3	3	**9 Medium**	R09	Instrument
N1.10	Composition	OTHER THAN	Impurity in the CO_2 or H_2	• Off-specification gas (sulphur or chloride in CO_2, moisture)	Slow poisoning of the Cu/ZnO/Al_2O_3 catalyst; activity and yield fall over days; no immediate hazard	• Supplier certificate (assumed) • AY-620 yield trend • GC	2	2	**4 Low**	R10	Operations

N2 Ethanol feed

Project	MTJA CO_2 methanol pilot · R-601	Node	N2 Ethanol feed
Design intent	Pump ethanol from T-301 with the P-301 metering pump at the design rate (FIC-301 on FT-301) into the gas feed ahead of E-601; the pressure-holding valve keeps PT-301 ≈ 55.7 bar g.	Drawing	2407-PE-PID-001 A6 (ethanol sheet); 2407-IN-CE-001 A2 rows 007–010, 015–020; 2407-PE-RLB-001 PSV-301/302
Conditions	Ethanol 68.2 mL/min at ≈ 21 °C; PT-301 55.7 bar g; T-301 at ≈ 0.2 bar g	Instruments	LT-301 (LAL/LAH-301), FIC-301 (FT-301), PT-301 (PAL/PAH, PALL/PAHH-301)
Equipment	T-301 ethanol tank (1.274 m³, design 2 bar g); P-301 metering pump (rated 17.8 L/h, max discharge 56 bar g); XV-301; check valves CK301/CK302; PSV-301 (T-301), PSV-302 (pump discharge)	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	NO	No ethanol flow	• P-301 motor trip or power dip • XV-301 closing • T-301 empty (pump runs dry) • suction strainer blocked	The bed dries: the wetted ester route stops and the evaporative cooling is lost; H_2 and CO_2 continue over a dry hot Cu bed; yield falls	• FAL-301 at 95 % of SP • PAL-301 53 bar g • PALL-301 trip at 50 bar g (2 s delay, armed in RUN only) • LAL-301 20 % • TAH-605 158 °C	2	4	**8 Medium**	R11 R30	Control, Instrument
N2.2	Flow	LESS	Less ethanol than set	• P-301 check valve passing (wear), speed rises to compensate • FT-301 reading high so FIC-301 cuts the real flow • gas in the pump head	EtOH:CO_2 below 4; wetting falls and the bed nears the dew point (≈ 152.5 °C); less ester route; FIC-301 output creeps up	• FAL-301 ±5 % of SP (same meter, cannot see a meter error) • FIC-301 output on the faceplate • LT-301 trend and the field weigh test • AY-620	2	3	**6 Medium**	R12	Operations
N2.3	Flow	MORE	More ethanol than set	• FIC-301 output failing high or left in MAN • set-point error • FY-100 ratio error	LHSV above the fitted range; the bed cools and yield per pass falls; D-601 fills faster; more liquid to D-602/T-601	• FAH-301 at 105 % of SP • FAHH-301 at 110 % of SP (C&E row 017, alarm only) • LAH-601 60 % • P-301 rated capacity 297 mL/min	2	3	**6 Medium**	—	—
N2.4	Pressure	MORE	High ethanol discharge pressure	• Blocked discharge: XV-301 closed with P-301 running, a manual valve closed, CK302 stuck • positive-displacement pump dead-heads	Pump discharge pressure rises quickly toward the pump maximum; leak or rupture of the discharge tubing; ethanol spray (flammable)	• PAH-301 59 bar g • PAHH-301 trip at 62 bar g (1 s delay, opens XV-603) • PSV-302 set 36 bar g (2407-PE-RLB-001, governing case blocked outlet), above the 27–32 bar g on/off band of P-301 on the plant HMI	3	3	**9 Medium**	R13	Process
N2.5	Flow	REVERSE	Gas back-flow into the ethanol system	• P-301 stopped with XV-301 open • check valves CK301/CK302 passing • reactor at 30 bar g	H_2 and CO_2 into the pump and T-301 (design 2 bar g): overpressure of T-301 and flammable gas in its vapour space	• Check valves CK301, CK302 • XV-301 fails closed and closes on every trip • PSV-301 2 bar g (sized for fire only)	4	2	**8 Medium**	R06 R14	Maintenance, Process
N2.6	Level	LESS	Low T-301 level	• Ethanol not replenished • transfer error • drain valve open	P-301 runs dry: flow lost (N2.1) and pump damage	• LAL-301 20 % • LT-301 indication • low level switch on T-301 in the instrument list (not on the C&E)	2	3	**6 Medium**	R11	Instrument
N2.7	Level	MORE	T-301 overfilled	• Transfer left running • level transmitter fault	Ethanol overflow and spill (flash point ≈ 13 °C); vapour cloud near electrical equipment; fire	• LAH-301 80 % • level switch LS-301 (instrument list) • operator present during transfer	3	3	**9 Medium**	R15	Operations
N2.8	Temperature	MORE	T-301 exposed to an external fire	• Pool fire from an ethanol spill or a nearby fire	Ethanol boils off; T-301 overpressure (design 2 bar g); escalation to the skid	• PSV-301 set 2 bar g, governing case external fire (543 kg/h in 2407-PE-RLB-001) • fire extinguishers (assumed)	4	2	**8 Medium**	R16	Safety
N2.9	Composition	OTHER THAN	Wrong liquid in T-301	• Wet ethanol, returned product or another solvent charged by mistake	Water inhibits the catalyst and shifts the route; yield falls; with another solvent unknown reactions	• Receipt check and sample (assumed) • AY-620 yield trend • lab analysis of D-602 liquid	2	2	**4 Low**	—	—

N3 E-601 electric preheater

Project	MTJA CO_2 methanol pilot · R-601	Node	N3 E-601 electric preheater
Design intent	Heat the mixed gas and ethanol to ≈ 127.0 °C two-phase at the R-601 inlet with the element held at 150 °C by TIC-601.	Drawing	2407-PE-PID-001 A6; 2407-IN-CE-001 A2 rows 027–032; 2407-ME-EQL-001
Conditions	Element 150.0 °C; outlet 127.0 °C; ≈ 30 bar g; duty 0.58 kW	Instruments	TIC-601 (TT-601, TAH-601, TAHH-601), TT-602 (TAL/TAH/TAHH-602), FT-601, JI-601 power (assumed tag)
Equipment	E-601 electric heater on the pipe (4.62 kW, 6 elements, design 36 bar g); element thermal cut-out TSC-601	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N3.1	Temperature	MORE	E-601 element and outlet too hot	• TIC-601 output failing high (SCR shorted, controller in MAN at high output) • TT-601 reading low	Element overheats; ethanol cracking and coke on the element; hot feed dries the top of the bed	• TAH-601 158 °C • TAHH-601 trip at 167 °C (2 s delay, opens XV-603) • TAHH-602 trip at 167 °C (2 s delay, opens XV-603) • TSC-601 element cut-out at 180 °C (model)	3	3	**9 Medium**	—	—
N3.2	Temperature	LESS	E-601 outlet too cold	• Elements failing (part or all of the 6) • TIC-601 failing low • TT-601 reading high	Colder liquid to the top of the bed, top-zone temperature and yield fall; no hazard	• TAL-602 110 °C (re-based for this unit, assumed • C&E has TAL 143 / TALL 135 °C) • JI-601 power indication	1	3	**3 Low**	R17	Process
N3.3	Flow	NO	No flow through E-601 with the heater on	• All feeds stopped (pack empty, pump stop) while TIC-601 keeps power on • trip that fails to cut the heater	Element temperature rises with no heat sink; local overheating of the pipe; coking	• TIC-601 cuts power as the element rises • TAH-601, TAHH-601 trip at 167 °C (2 s delay, opens XV-603) • TSC-601 cut-out • heaters off on every trip	3	2	**6 Medium**	—	—
N3.4	Pressure	MORE	E-601 / R-601 exposed to fire	• External fire in the skid area (ethanol spill, H_2 jet fire)	Liquid in E-601 and R-601 vaporises; overpressure of E-601 and R-601 (design 36 bar g)	• PSV-601 36 bar g, governing case external fire with liquid vaporisation (400.1 kg/h) • no block valve between E-601 and R-601	4	2	**8 Medium**	R16	Safety
N3.5	Composition	OTHER THAN	H_2 leak at a hot flange or fitting	• Thermal cycling of the E-601 and R-601 connections • gasket failure • fitting not tightened after maintenance	H_2-rich gas at 30 bar g and ≈ 150 °C escapes; jet fire or explosion in the skid; flame almost invisible	• Leak test before start-up (procedure, assumed) • PAL-601 29 bar g and PALL-601 trip at 27 bar g (2 s delay, armed in RUN only) for a large leak only • gas detectors GD-001/GD-002 read ppm on the plant HMI (alarm to the operator • type, location and set points not in the documents) • no automatic trip on gas detection (no detector row on the C&E)	5	2	**10 High**	R18	Safety
N3.6	Reaction	AS WELL AS	Fouling on the heated surface	• Ethanol decomposition and deposits on the element	More element temperature for the same duty; earlier TAH-601; uneven heating	• TIC-601 output trend • TAH-601 158 °C	2	2	**4 Low**	—	—

N4 R-601 trickle bed and band heaters

Project	MTJA CO_2 methanol pilot · R-601	Node	N4 R-601 trickle bed and band heaters
Design intent	Keep the 17.34 kg Cu/ZnO/Al_2O_3 bed at ≈ 130–151 °C, below the ≈ 152.5 °C dew point so the bed stays wetted, at ≈ 30 bar g, with the three band-heater zones E1/E2/E3 on wall control (TIC-611/612/613); TIC-604 cascade normally in MAN.	Drawing	2407-PE-PID-001 A6 (reactor sheet); 2407-IN-CE-001 A2 rows 021–024, 033–034; 2407-ME-EQL-001
Conditions	Bed TT-603/604/605 130.2 / 148.3 / 151.4 °C; walls 180 / 162 / 152 °C (assumed); heater duty 2.37 / 2.04 / 1.72 kW	Instruments	TT-603, TT-604, TT-605 (TAH/TAHH-605); TIC-611/612/613 wall (assumed tags); TIC-604 cascade master; JI-611/612/613 (assumed); PT-601
Equipment	R-601 (73.66 mm × 3.23 m bed, design 36 bar g / 180 °C); band heaters E1/E2/E3 (assumed 4 kW each, SCR)	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N4.1	Temperature	MORE	Middle zone too hot	• E2 band-heater SCR failing on at full power • TIC-612 output failing high	Hot spot in the middle of the bed far above the dew point: the bed dries, the ester route stops, Cu sinters (permanent loss of activity); tube metal above the 180 °C design temperature at 30 bar g with H_2: loss of strength, possible loss of containment and H_2 fire	• TAH-605 158 °C and TAHH-605 trip at 167 °C (2 s delay, opens XV-603) on the bottom TC only • TT-604 has no alarm • TIC-612 wall high alarm 260 °C (assumed) • band heaters off on a trip (assumed) • PSV-601 36 bar g	4	3	**12 High**	R19 R20	Instrument, Process
N4.2	Temperature	MORE	Top zone too hot	• E1 band-heater SCR or TIC-611 output failing high • TIC-611 in MAN at a high output	Hot spot in the top zone below TT-603; bed dries and sinters; wall far above the design temperature; the bottom TC and TT-603 hardly move, so nothing alarms	• TIC-611 wall high alarm 260 °C (assumed) • TT-603 indication (no alarm) • TAHH-605 trip at 167 °C (2 s delay, opens XV-603) (bottom only)	4	3	**12 High**	R19 R20 R21	Instrument, Process
N4.3	Temperature	MORE	Bed hotter than indicated in cascade	• TT-604 reading low while TIC-604 is in cascade on TIC-612 (thermocouple drift, junction moved) • wrong cascade set point	TIC-604 raises the E2 wall to correct a false low reading; the real mid-bed goes above the dew point and the walls well above design; no alarm	• TIC-604 normally in MAN • TT-603/TT-605 neighbours on the display • TAH-605 158 °C	3	3	**9 Medium**	R19 R22	Control, Instrument
N4.4	Temperature	LESS	Zone too cold	• E1 SCR failing off or a heater element open circuit • insulation damaged • TIC-611 reading high	Top of the bed cools, more liquid hold-up, yield falls; no hazard	• JI-611 power indication (assumed tag) • TT-603 trend • AY-620 yield	1	3	**3 Low**	R19	Instrument
N4.5	Reaction	PART OF	Only part of the bed wetted	• Liquid channelling or a poor distributor • low ethanol rate • tilt of the tube	Dry zones run hotter and lose the ester route; yield falls; local hot spots between the thermocouples	• TT-603/604/605 profile • AY-620 yield soft sensor • liquid distributor (design)	2	3	**6 Medium**	R21	Instrument
N4.6	Pressure	LESS	R-601 pressure low	• BPV-601 drifting low (spring relaxing) • BPV set wrong • leak	Dew point falls with pressure, so the bed dries; yield falls; larger vent flow	• PAL-601 29 bar g • PALL-601 trip at 27 bar g (2 s delay, armed in RUN only)	2	3	**6 Medium**	R30	Control
N4.7	Reaction	OTHER THAN	Reduced catalyst exposed to air	• Air ingress while the bed is warm after a shutdown or during unloading • N_2 blanket lost	Pyrophoric re-oxidation of reduced Cu: fast temperature rise, catalyst damage, ignition of residual ethanol	• Shutdown procedure cools under N_2 to TT-605 < 60 °C before depressuring • XV-401 fails open	4	2	**8 Medium**	R29 R38	Process

N5 F-601 filter, E-602 condenser and chilled water

Project	MTJA CO_2 methanol pilot · R-601	Node	N5 F-601 filter, E-602 condenser and chilled water
Design intent	Filter the reactor outlet (F-601) and condense the ethanol, methanol, water and esters in E-602 with chilled water at 15 °C, 67 L/min, so the condensate leaves at ≈ 30.4 °C to D-601.	Drawing	2407-PE-PID-001 A6; 2407-IN-CE-001 A2 rows 045–049; 2407-PE-RLB-001 PSV-602
Conditions	Process 30 bar g; condensate 30.4 °C; chilled water 15 °C at 67 L/min	Instruments	FT-701 (FAL-701, FALL-701), TT-609 (TAL/TAH/TAHH-609), PT-601 and PT-608 either side of F-601
Equipment	F-601 filter; E-602 shell-and-tube condenser (4 kW; shell design 10 bar g in the equipment list, PSV-602 set 4 bar g); chiller, chilled-water pump and 150 L tank	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	NO	No chilled-water flow	• Chilled-water pump trip • valve closed • power loss	Hot vapour reaches D-601: condensables go to the cold vent with the gas (flammable vapour release), D-601 heats up, product lost	• FAL-701 63.65 L/min • FALL-701 trip at 60.3 L/min (5 s delay, armed in RUN only, opens XV-603) • TAH-609 32 °C • TAHH-609 trip at 34 °C (5 s delay, opens XV-603)	3	3	**9 Medium**	R30	Control
N5.2	Flow	LESS	Less chilled-water flow	• Strainer fouling • valve partly closed • pump wear	TT-609 rises slowly; more vapour in the vent gas	• FAL-701 63.65 L/min and FALL-701 trip at 60.3 L/min (5 s delay, armed in RUN only, opens XV-603): a narrow band below the 67 L/min design • TAH-609	2	3	**6 Medium**	—	—
N5.3	Temperature	MORE	Chilled water too warm	• Chiller compressor trip with the pump still running • chiller undersized on a hot day	Condensate temperature climbs slowly; methanol and ethanol lost to the vent; no trip for a long time	• TAH-609 32 °C • TAHH-609 trip at 34 °C (5 s delay, opens XV-603) • no chiller-status or chilled-water supply temperature alarm	2	4	**8 Medium**	R24	Instrument
N5.4	Temperature	LESS	Condensate too cold	• Chiller set point too low • low load at reduced rate	Ice or CO_2 hydrate can form at 30 bar g with the water in the condensate: blockage at the E-602 outlet	• TAL-609 8 °C	2	2	**4 Low**	—	—
N5.5	Pressure	MORE	Process gas into the chilled-water side	• E-602 tube leak or rupture (30 bar g tube side into the low-pressure shell)	H_2 and CO_2 into the chilled-water loop; shell and chiller tank overpressure; flammable gas released at the tank	• PSV-602 set 4 bar g, governing case tube rupture (sized by the vendor, 2407-PE-RLB-001)	4	2	**8 Medium**	R25	Process
N5.6	Flow	LESS	Restricted flow through F-601	• Filter plugging with catalyst fines or coke	Pressure drop rises: PT-601 rises against PT-608; filter element may collapse and release fines downstream	• PAH-601 32 bar g • PAHH-601 trip at 33 bar g (1 s delay, opens XV-603) • no differential-pressure alarm across F-601	2	3	**6 Medium**	R23	Instrument

N6 D-601 HP separator, BPV-601, vent and GC

Project	MTJA CO_2 methanol pilot · R-601	Node	N6 D-601 HP separator, BPV-601, vent and GC
Design intent	Separate the condensate from the gas at 30 bar g; hold the pressure with BPV-601 and send ≈ 37.4 NL/min of gas to the cold vent past the GC; drain the liquid on/off to D-602 by LIC-601 through XV-601; XV-603 depressures to the cold vent.	Drawing	2407-PE-PID-001 A6 (separator sheet); 2407-IN-CE-001 A2 rows 025–026, 035–038, 054; 2407-PE-RLB-001 PSV-601/603
Conditions	PT-608 30.09 bar g; 30.4 °C; vent H_2 84.4 %, CO_2 14.9 %, CO 0.66 % (dry)	Instruments	PT-608 (PAH/PAHH-608), LIC-601 (LT-601, LAL/LAH-601, LALL/LAHH-601), LS-605 (LAHH-605), FT-603, GC AI-610/611/612/613
Equipment	D-601 (0.061 m^3, design 36 bar g / 75 °C); BPV-601; XV-601 (FC), XV-603 (FO, BDV); PSV-601, PSV-603; cold vent; GC	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N6.1	Pressure	MORE	D-601 and R-601 pressure high	• BPV-601 sticking closed (seat fouled) • vent line blocked (ice, closed valve) • BPV set too high	Pressure rises in the whole HP loop toward the 36 bar g design; PSV-601/603 lift to the vent; flange leaks of H_2	• PAH-601/PAH-608 32 bar g • PAHH-601 trip at 33 bar g (1 s delay, opens XV-603) • PAHH-608 trip at 33 bar g (1 s delay, opens XV-603) • PSV-601 36 bar g • PSV-603 36 bar g	4	2	**8 Medium**	—	—
N6.2	Pressure	LESS	D-601 pressure low	• BPV-601 spring relaxing or set low • XV-603 opened in error or on air loss (fails open) • PSV passing	Depressure of the loop; the bed dries; flashing in D-601; a high gas rate to the vent	• PAL-601 29 bar g • PALL-601 trip at 27 bar g (2 s delay, armed in RUN only)	2	3	**6 Medium**	R30	Control
N6.3	Level	MORE	D-601 level high	• XV-601 stuck closed • LT-601 frozen at its last value • LIC-601 in MAN	Liquid carried into BPV-601, the vent line and the GC sample; ethanol liquid at the cold vent; the level rises slowly because the liquid make is small, so the alarm comes long after the cause	• LAH-601 60 % • LAHH-601 trip at 80 % (5 s delay, opens XV-603) • LAHH-605 trip at 85 % (5 s delay, opens XV-603) on the independent level switch LS-605 • sight glass LG-601	3	3	**9 Medium**	R27	Instrument
N6.4	Level	LESS	D-601 level low or empty	• XV-601 passing or held open • LIC-601 switching levels wrong • LT-601 reading high	D-601 empties and HP gas blows through XV-601 into D-602 (design 6 bar g)	• LAL-601 20 % • LALL-601 trip at 10 % (5 s delay, armed in RUN only, opens XV-603) • the same XV-601 is the control and the trip valve	3	3	**9 Medium**	R27 R30	Control, Instrument
N6.5	Composition	AS WELL AS	Flammable and toxic gas at the vent	• Normal operation: H_2-rich vent gas with CO and solvent vapour • more during a depressure through XV-603 or a PSV lift	Flammable plume at the vent outlet (ignition by lightning or static: vent fire); CO exposure near the vent and the GC	• Cold vent to a safe location (assumed) • FT-603 indication • GC AI-612 CO	4	2	**8 Medium**	R26	Safety
N6.6	Composition	OTHER THAN	GC reports a wrong vent composition	• Wrong calibration factor for CO • sample line wet or blocked	CO and yield reported wrong; CO exposure underestimated; wrong decisions on the feed ratio	• Calibration routine (assumed) • AY-620 soft sensor as a cross-check	1	3	**3 Low**	R28	Instrument
N6.7	Flow	REVERSE	Air drawn back from the vent	• Unit depressured and cooling with the N_2 stopped (gas contracts) • vent open to atmosphere	Air into D-601, E-602 and R-601: O_2 meets residual H_2 and reduced catalyst (see N4.7)	• BPV-601 closes below its set pressure • N_2 blanket in the shutdown procedure (assumed)	4	1	**4 Low**	R29	Process
N6.8	Pressure	MORE	D-601 exposed to fire	• External fire in the separator area	Gas expansion and liquid vaporisation in D-601; overpressure	• PSV-603 set 36 bar g, governing case external fire (294.4 kg/h in 2407-PE-RLB-001)	4	2	**8 Medium**	R16	Safety

N7 D-602 LP flash and T-601 product

Project	MTJA CO_2 methanol pilot · R-601	Node	N7 D-602 LP flash and T-601 product
Design intent	Flash the D-601 liquid at ≈ 0.2 bar g in D-602 (BPV-602), drain it on/off by LIC-602 through XV-602 to the atmospheric product tank T-601; P-302 transfers product.	Drawing	2407-PE-PID-001 A6; 2407-IN-CE-001 A2 rows 039–044; 2407-PE-RLB-001 PSV-604/606
Conditions	D-602 ≈ 0.2 bar g, ≈ 30 °C; T-601 atmospheric	Instruments	LIC-602 (LT-602, LAL/LAH-602, LALL/LAHH-602), T-601 level (LAH/LAL-603)
Equipment	D-602 (0.061 m³, design 6 bar g / 75 °C); BPV-602; XV-602 (FC); PSV-606 (2 bar g); T-601 product tank (0.1 m³); PSV-604; P-302; T-602 waste IBC	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N7.1	Pressure	MORE	Gas blow-by into D-602	• D-601 drained with XV-601 open (passing, LIC-601 in MAN) while the low-low trips are bypassed in READY at start-up	HP gas into the 6 bar g D-602: overpressure, PSV-606 lifts; flammable gas to T-601 and the area	• LALL-601 trip at 10 % (5 s delay, armed in RUN only, opens XV-603) (armed in RUN only) • LAHH-602 trip at 80 % (5 s delay) (armed always) • BPV-602 • PSV-606 2 bar g (relief cases blow-by 22.72 kg/h and fire)	3	3	**9 Medium**	R30	Control
N7.2	Level	MORE	D-602 level high	• XV-602 stuck closed • LIC-602 in MAN • T-601 back-pressure	Liquid backs up; D-602 floods into its vent; D-601 cannot drain	• LAH-602 60 % • LAHH-602 trip at 80 % (5 s delay) opens XV-602	2	3	**6 Medium**	—	—
N7.3	Level	LESS	D-602 empty	• XV-602 passing or held open	Gas from D-602 to the atmospheric T-601; solvent vapour in the tank area	• LAL-602 20 % • LALL-602 trip at 10 % (5 s delay, armed in RUN only)	3	2	**6 Medium**	R30	Control
N7.4	Level	MORE	T-601 overfilled	• P-302 not started • level indication wrong • long run without transfer	Methanol and ethanol spill (toxic and flammable) around T-601	• LAH-603 80 % (C&E row 043, alarm only) • level switches LS-603/604 (instrument list) • PSV-604	3	3	**9 Medium**	R31	Instrument
N7.5	Composition	OTHER THAN	Operator exposed to methanol	• Open sampling or drumming of the product • leak at the P-302 connection	Methanol vapour and skin contact (toxic); ethanol fire risk	• PPE and sampling practice (assumed)	3	3	**9 Medium**	R32	Operations
N7.6	Temperature	MORE	T-601 exposed to fire	• Pool fire in the product area	Product boils off; T-601 overpressure	• PSV-604 set 2 bar g, governing case external fire (179.8 kg/h in 2407-PE-RLB-001)	3	2	**6 Medium**	R16	Safety

N8 Utilities: instrument air and electrical power

Project	MTJA CO_2 methanol pilot · R-601	Node	N8 Utilities: instrument air and electrical power
Design intent	Supply instrument air at ≈ 4.0 bar g to the XVs and the pressure-holding valve, and power to the heaters, P-301, chiller, chilled-water pump, MFCs, GC and the SIS/DCS.	Drawing	2407-PE-PID-001 A6 (utility sheet); 2407-IN-CE-001 A2 rows 050–053; 2407-EE-LL-001
Conditions	Air header 4.0 bar g; the valves move to their fail positions below ≈ 2 bar g (model)	Instruments	PT-902 (PAL/PAH-902, PALL/PAHH-902)
Equipment	Air compressor, PR-901, air header; electrical supply and distribution (UPS not shown in the documents)	Team / date	

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N8.1	Air / power	NO	No instrument air	• Air compressor trip • header leak • PR-901 failing closed	All air-operated valves go to their fail positions: feeds and drains close, XV-401 opens the N_2 purge and XV-603 depressures; the unit trips on the first transmitter that reacts, which may not be the air header	• PAL-902 1.2 bar g • PALL-902 trip at 0.9 bar g (2 s delay) • fail positions: XV-101/201/301 fail closed, XV-401 and FV-401 fail open, XV-601/602 fail closed, XV-603 fails open	2	3	**6 Medium**	R33	Instrument
N8.2	Air / power	MORE	Instrument-air pressure high	• PR-901 failing open	Actuators and SOVs above their rating	• PAH-902 4.8 bar g • PAHH-902 trip at 5.1 bar g (2 s delay)	1	2	**2 Low**	—	—
N8.3	Air / power	NO	No electrical power	• Site supply failure • main breaker trip	Heaters, P-301, chiller and chilled-water pump stop together; the SIS de-energises to trip; if the SIS, MFC or GC power is lost there is no indication of the state of the unit	• De-energise-to-trip SIS (assumed) • fail-safe valves (fail positions: XV-101/201/301 fail closed, XV-401 and FV-401 fail open, XV-601/602 fail closed, XV-603 fails open) • N_2 purge from the packs needs no power	3	3	**9 Medium**	R34	Electrical
N8.4	Air / power	LESS	Voltage dip	• Grid disturbance	Motors (P-301, chilled-water pump) drop out while the heaters ride through: loss of ethanol or cooling with the heat on	• PALL-301 trip at 50 bar g (2 s delay, armed in RUN only) • FALL-701 trip at 60.3 L/min (5 s delay, armed in RUN only, opens XV-603) • TAH-605 158 °C	2	3	**6 Medium**	R34	Electrical

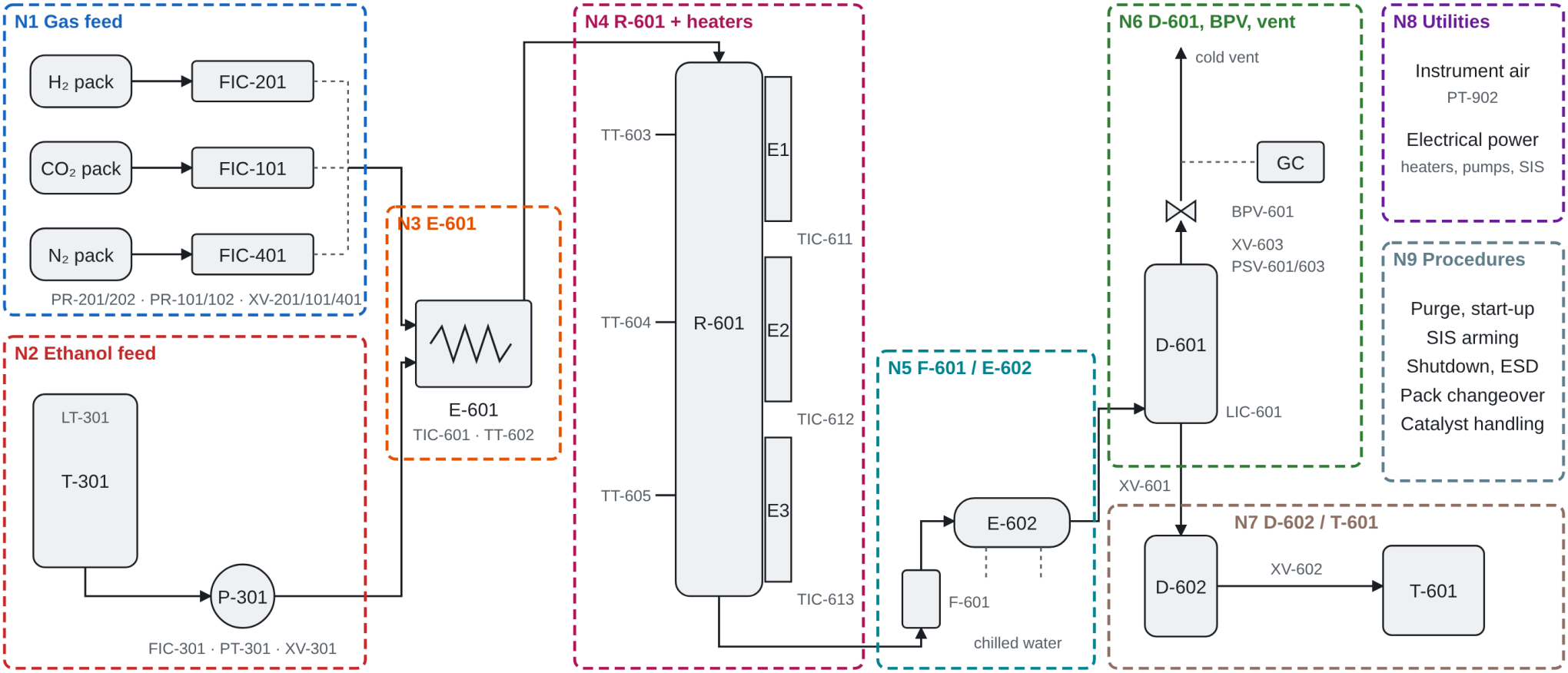
N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst

Project	MTJA CO_2 methanol pilot · R-601	Node	N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst
Design intent	Take the unit from cold, depressured and N_2-filled to the design point through the SIS states TRIPPED → READY → RUN; shut it down under N_2; change gas packs; load, reduce and unload the catalyst (procedures for the last are not in the unit documents).	Drawing	OTS procedures: cold start-up, normal shutdown, emergency shutdown, pack changeover; 2407-IN-CE-001 A2
Conditions	Start-up permissives: R-601 inlet pressure PT-601 ≥ 29 bar g; P-301 running, PT-301 ≥ 53 bar g; Chilled water FT-701 ≥ 63.7 L/min; D-601 level LT-601 ≥ 20 %; D-602 level LT-602 ≥ 20 %; H_2 and CO_2 supply ≥ 27 bar g; XV-603 depressure valve closed	Instruments	SIS state, permissives, first-out; all transmitters
Equipment	Whole unit; SIS (ESD-SW, reset, arm); N_2 pack; gas packs	Team / date	

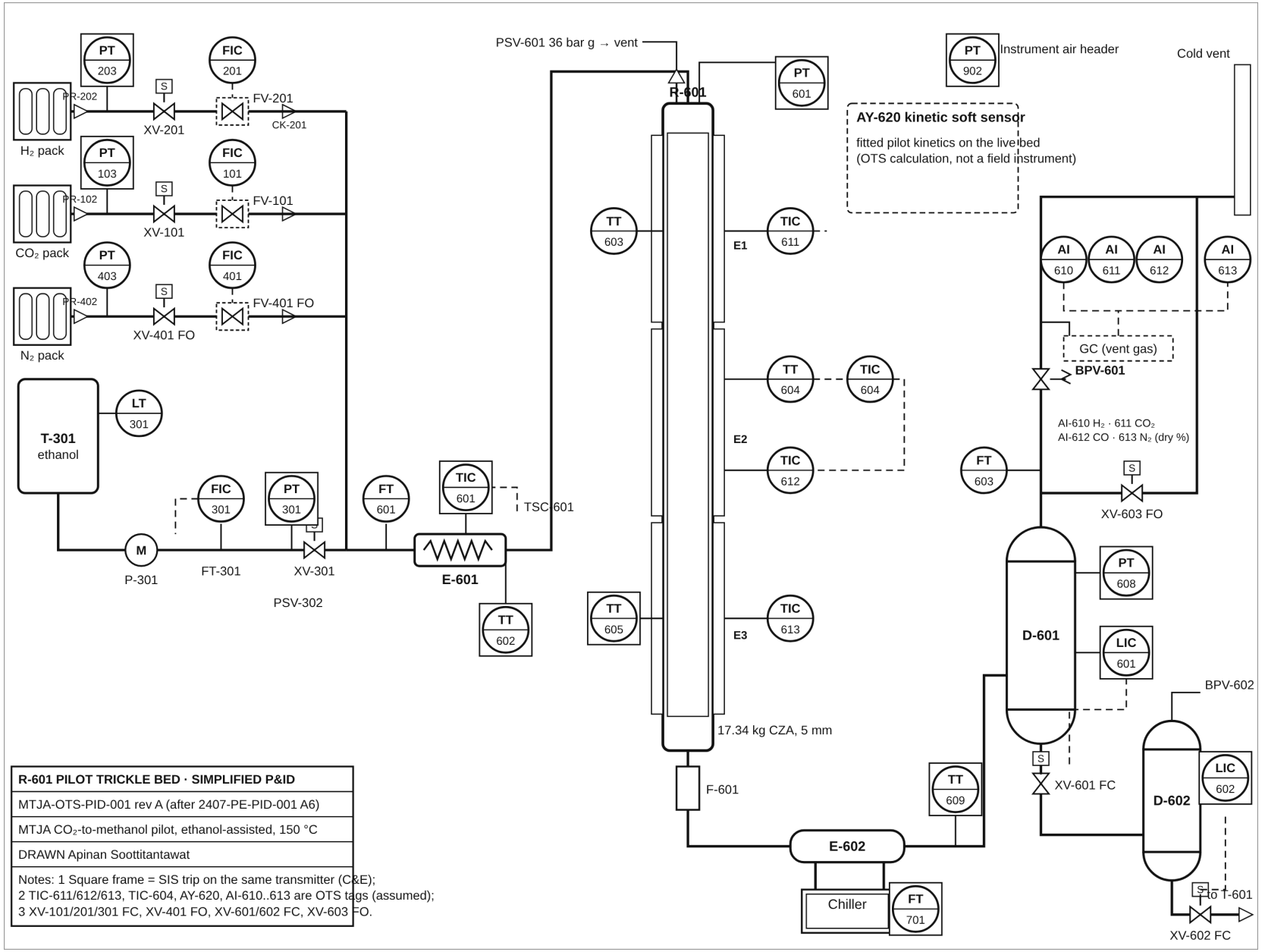
ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N9.1	Sequence	BEFORE	H_2 admitted before the air is purged out	• Start-up after maintenance with the unit only pressurised with N_2, not purged to a low O_2 content • step skipped	H_2/air flammable mixture in R-601 over hot catalyst: explosion or fast exotherm	• Start-up procedure pressurises with N_2 to PT-601 ≥ 29 bar g before heating • no O_2 measurement	5	2	**10 High**	R35	Operations
N9.2	Sequence	EARLY	H_2 in before the ethanol flows	• Steps done out of order • P-301 not started or not delivering	H_2 and CO_2 over a dry hot Cu bed: no evaporative cooling, hot spots, catalyst damage	• Start-up step order (ethanol, then H_2, then CO_2) • TAH-605 158 °C • PALL-301 bypassed in READY	3	3	**9 Medium**	R36	Control
N9.3	Time	LATE	SIS left in READY	• Arming forgotten after line-out • permissive not met and nobody follows up	The low-low trips (PALL-301, PALL-601, LALL-601, LALL-602, FALL-701) stay bypassed while the unit runs	• Start-up procedure step "Arm the SIS (RUN)" • SIS state on the DCS	3	3	**9 Medium**	R30	Control
N9.4	Time	EARLY	SIS armed too early	• Arming before the flows and levels have settled	Spurious trip on a low-low; restart and the hazards of a second start-up	• Permissives to arm: R-601 inlet pressure PT-601 ≥ 29 bar g; P-301 running, PT-301 ≥ 53 bar g; Chilled water FT-701 ≥ 63.7 L/min; D-601 level LT-601 ≥ 20 %; D-602 level LT-602 ≥ 20 %; H_2 and CO_2 supply ≥ 27 bar g; XV-603 depressure valve closed	1	4	**4 Low**	—	—
N9.5	Sequence	BEFORE	Depressured before the bed is cooled	• XV-603 opened with the bed hot and full of liquid	Ethanol flashes and is carried to the vent; vent flow and flammable plume larger than needed	• Shutdown procedure order: feeds off, heaters off, cool under N_2 to TT-605 < 60 °C, then depressure	3	3	**9 Medium**	—	—
N9.6	Sequence	AFTER	Heaters left on after the feeds stop	• Shutdown steps out of order • heater switched back on	Dry bed and empty E-601 heated: wall and element above design temperature	• Shutdown step "Heaters off" • TAH-601 • TIC-611/612/613 wall alarms (assumed) • heaters trip with the SIS	3	2	**6 Medium**	R20	Process
N9.7	Time	LATE	H_2 pack changeover late	• Low-pressure alarm not acted on in time	PALL-203 trips the unit; production lost; restart	• PAL-203 27 bar g • PALL-203 trip at 24 bar g (2 s delay) • spare pack	1	4	**4 Low**	R02	Process
N9.8	Composition	OTHER THAN	Air in the H_2 manifold after a changeover	• Pigtail not vent-purged after connecting the new pack	Air slug with H_2 to the reactor (see N1.8)	• Pack-change practice (assumed)	4	2	**8 Medium**	R37	Operations
N9.9	Maintenance	OTHER THAN	Catalyst loading, reduction or unloading done wrongly	• No written procedure in the unit documents • reduction with too much H_2 or too fast • unloading while the catalyst is reduced and warm	Reduction exotherm sinters the catalyst; pyrophoric catalyst on unloading; dust exposure	• None documented	4	2	**8 Medium**	R38	Process
N9.10	Maintenance	BEFORE	Equipment opened before it is gas-free	• Isolation and purge incomplete • residual H_2 or ethanol in a dead leg	Flammable or toxic release at the opened joint; fire	• Permit to work (assumed)	4	2	**8 Medium**	R39	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Safeguards	S	L	Risk	Rec.	Action party
N9.11	Time	LATE	Emergency shutdown late	• Operator hesitates • ESD-SW not reachable	An escalating event (leak, fire) is fed longer	• ESD-SW (C&E row 001) with the full trip action: closes XV-101/201/301 and FV-101/201, stops P-301, opens XV-401 and FV-401 (N_2 purge), E-601 and the band heaters off • XV-603 opens	2	3	**6 Medium**	R40	Operations

Appendix 2 Drawings



Drawing A2-1. HAZOP node map of R-601.



Drawing A2-2. Simplified P&ID of R-601 on the simulator (after 2407-PE-PID-001 A6), the drawing the nodes were marked on. The issued P&ID 2407-PE-PID-001 A6 is the reference.

Appendix 3 Action close-out sheets

One sheet per recommendation (40 sheets). The response, status and sign-off are completed by the action party and the HAZOP leader.

Action close-out sheet R01

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R01
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Instrument / Control
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	6 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.1	More H_2 flow than set	FIC-201 MFC or its output failing high; controller left in MAN at a high output; set-point entry error; FY-100 ratio set wrong	H_2:CO_2 well above 5; more gas through BPV-601 to the cold vent (larger H_2-rich vent plume); bed temperature rises a few degrees; operation outside the fitted kinetic window	FAH-201 at 105 % of SP (only while FIC-201 is in AUTO); PAH-601 32 bar g; TAH-605 158 °C; MFC full scale 100 NL/min limits the flow; BPV-601 holds the pressure	**6 Medium**

Recommendation

Add an absolute high-flow alarm on FT-201 (and FT-101) that works in every controller mode, and an H_2:CO_2 ratio deviation alarm on FY-100.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R02

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R02
Node	N1 H_2, CO_2 and N_2 gas feed, N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Process / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.2	Less or no H_2 flow	H_2 pack running down; PR-202 failing closed; XV-201 closing (solenoid or air fault); MFC failing low; filter F201 plugging	H_2:CO_2 falls, methanol yield falls while the ester route continues; MFC loses its pressure drop before any alarm because PAL-203 is below the 30 bar g reactor pressure; if the supply falls below the reactor, back-flow (see N1.6)	FAL-201 at 95 % of SP; PAL-203 27 bar g; PALL-203 trip at 24 bar g (2 s delay); spare H_2 pack (PACK B) with changeover in the field	**8 Medium**
N1.4	Low CO_2 supply pressure	CO_2 pack cold (cylinder cooling during draw-off), pack empty, regulator PR-101 icing	CO_2 flow falls, H_2:CO_2 rises; regulator icing can make the pressure swing; loss of CO_2 trips the unit	PAL-103 27 bar g; PALL-103 trip at 24 bar g (2 s delay, opens XV-603); FAL-101 at 95 % of SP	**8 Medium**
N9.7	H_2 pack changeover late	Low-pressure alarm not acted on in time	PALL-203 trips the unit; production lost; restart	PAL-203 27 bar g; PALL-203 trip at 24 bar g (2 s delay); spare pack	**4 Low**

Recommendation

Raise PAL-203 and PAL-103 above the reactor pressure plus the MFC pressure drop (now 27 bar g, below the 30 bar g reactor), and review PALL-203/103 against the pressure at which the MFCs lose control.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R03

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R03
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Operations / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	6 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.3	Less real CO_2 than indicated	FT-101 calibration drifting high, so FIC-101 cuts the real CO_2; MFC failing low; PR-102 freezing	H_2:CO_2 above 5 unseen by the operator; yield per CO_2 and the reported conversion are wrong; the FAL/FAH-101 deviation alarms use the same transmitter and cannot see the error	FAL/FAH-101 ±5 % of SP (same transmitter); GC AI-611 CO_2 in the vent gas every 240 s; AY-620 yield soft sensor; PAL-103 27 bar g	**6 Medium**

Recommendation

Check FT-101 and FT-201 against a reference meter each campaign and reconcile the CO_2 feed with a GC carbon balance.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R04

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R04
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Process / Mechanical
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.4	Low CO_2 supply pressure	CO_2 pack cold (cylinder cooling during draw-off), pack empty, regulator PR-101 icing	CO_2 flow falls, H_2:CO_2 rises; regulator icing can make the pressure swing; loss of CO_2 trips the unit	PAL-103 27 bar g; PALL-103 trip at 24 bar g (2 s delay, opens XV-603); FAL-101 at 95 % of SP	**8 Medium**

Recommendation

Protect the CO_2 pack regulator against cooling and icing (heated or insulated PR-101) and fit a switch-over manifold.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R05

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R05
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Process
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.5	Full pack pressure downstream of the regulators	First and second-stage regulator (PR-202, PR-102 or PR-402) failing open or set wrong	MFCs, XVs and tubing see full pack pressure; R-601 pressure rises if the MFC passes; overpressure of components rated for the regulated pressure; H_2 release at a fitting	Two regulators in series; PAH-601 32 bar g; PAHH-601 trip at 33 bar g (1 s delay, opens XV-603); PSV-601 36 bar g on E-601/R-601 (no relief valve on the feed manifold in 2407-PE-RLB-001)	**8 Medium**

Recommendation

Confirm the pressure rating of the MFCs, XVs and tubing downstream of PR-202, PR-102 and PR-402 against full pack pressure, or add relief valves there (none in 2407-PE-RLB-001).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R06

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R06
Node	N1 H_2, CO_2 and N_2 gas feed, N2 Ethanol feed	Action party	Maintenance
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.6	Back-flow from the reactor into a gas supply	Supply pressure below the reactor (empty pack with XV open; CO_2 supply drawn down); MFC open during a pressure test	Reactor gas and ethanol vapour into the CO_2 or N_2 manifold and packs; flammable gas in the CO_2 system; contamination of the pack	Check valves CK201, CK101, CK401; PALL-203 trip at 24 bar g (2 s delay); PALL-103 trip at 24 bar g (2 s delay, opens XV-603); the trips close XV-201/101, which also fail closed	**6 Medium**
N2.5	Gas back-flow into the ethanol system	P-301 stopped with XV-301 open; check valves CK301/CK302 passing; reactor at 30 bar g	H_2 and CO_2 into the pump and T-301 (design 2 bar g): overpressure of T-301 and flammable gas in its vapour space	Check valves CK301, CK302; XV-301 fails closed and closes on every trip; PSV-301 2 bar g (sized for fire only)	**8 Medium**

Recommendation

Put the check valves CK101, CK201, CK301, CK302, CK401 and CK601 in the inspection and proof-test plan.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R07

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R07
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	3 Low

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.7	N_2 in the feed during operation	XV-401 or FV-401 passing; FIC-401 left with a set point after start-up	N_2 dilutes the feed, H_2 partial pressure and yield fall slowly; vent rate rises; no safety consequence	FT-401 indication; GC AI-613 N_2 in the vent gas; start-up step "stop the N_2"	3 Low

Recommendation

Add a high alarm on FT-401 while the SIS is in RUN (N_2 passing into the feed).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R08

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R08
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	5 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.8	Wrong gas connected to a feed line	Wrong pack or cylinder connected at changeover (air, O_2 or another gas on the H_2 or CO_2 line); pigtail not purged	Air or O_2 with H_2 over hot reduced Cu catalyst: flammable mixture, fast exotherm, catalyst oxidation, possible explosion in R-601	Gas-specific pack connections (assumed); pack labels; GC on the vent gas (after the event)	**5 Medium**

Recommendation

Use gas-specific pack connections and labels, with a two-person check at every pack change.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R09

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R09
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Instrument / Control
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.9	Low N_2 supply pressure	N_2 pack running down unnoticed; PR-402 failing closed	The N_2 purge that every trip relies on is not available: H_2 stays in R-601 after a trip; no inert for a shutdown	PAL-403 24 bar g; XV-401 and FV-401 fail open (purge path open on a trip). N_2 is not among the start-up permissives	**9 Medium**

Recommendation

Add the N_2 supply (PT-403 above PAL-403) to the start-up permissives, because every trip relies on the N_2 purge.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R10

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R10
Node	N1 H_2, CO_2 and N_2 gas feed	Action party	Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	4 Low

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.10	Impurity in the CO_2 or H_2	Off-specification gas (sulphur or chloride in CO_2, moisture)	Slow poisoning of the $Cu/ZnO/Al_2O_3$ catalyst; activity and yield fall over days; no immediate hazard	Supplier certificate (assumed); AY-620 yield trend; GC	4 Low

Recommendation

Check the supplier certificate (sulphur, chloride, moisture) for every CO_2 and H_2 pack received.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R11

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R11
Node	N2 Ethanol feed	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.1	No ethanol flow	P-301 motor trip or power dip; XV-301 closing; T-301 empty (pump runs dry); suction strainer blocked	The bed dries: the wetted ester route stops and the evaporative cooling is lost; H_2 and CO_2 continue over a dry hot Cu bed; yield falls	FAL-301 at 95 % of SP; PAL-301 53 bar g; PALL-301 trip at 50 bar g (2 s delay, armed in RUN only); LAL-301 20 %; TAH-605 158 °C	**8 Medium**
N2.6	Low T-301 level	Ethanol not replenished; transfer error; drain valve open	P-301 runs dry: flow lost (N2.1) and pump damage	LAL-301 20 %; LT-301 indication; low level switch on T-301 in the instrument list (not on the C&E)	**6 Medium**

Recommendation

Confirm that the T-301 low-level switch stops P-301 (dry running), and add it to the C&E.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R12

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R12
Node	N2 Ethanol feed	Action party	Operations / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	6 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.2	Less ethanol than set	P-301 check valve passing (wear), speed rises to compensate; FT-301 reading high so FIC-301 cuts the real flow; gas in the pump head	EtOH:CO_2 below 4; wetting falls and the bed nears the dew point (≈ 152.5 °C); less ester route; FIC-301 output creeps up	FAL-301 ±5 % of SP (same meter, cannot see a meter error); FIC-301 output on the faceplate; LT-301 trend and the field weigh test; AY-620	**6 Medium**

Recommendation

Reconcile the ethanol used daily (T-301 level and the weigh test against the FT-301 totaliser) and add a high-output alarm on FIC-301.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R13

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R13
Node	N2 Ethanol feed	Action party	Process / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.4	High ethanol discharge pressure	Blocked discharge: XV-301 closed with P-301 running, a manual valve closed, CK302 stuck; positive-displacement pump dead-heads	Pump discharge pressure rises quickly toward the pump maximum; leak or rupture of the discharge tubing; ethanol spray (flammable)	PAH-301 59 bar g; PAHH-301 trip at 62 bar g (1 s delay, opens XV-603); PSV-302 set 36 bar g (2407-PE-RLB-001, governing case blocked outlet), above the 27–32 bar g on/off band of P-301 on the plant HMI	**9 Medium**

Recommendation

Re-set the P-301 discharge alarms and trips to the plant basis: the plant HMI runs P-301 on/off on PT-301 at 27–32 bar g, which is consistent with PSV-302 at 36 bar g (2407-PE-RLB-001), but the C&E (PALL-301 50, PAHH-301 62 bar g) and the OTS model (≈ 56 bar g behind a pressure-holding valve) follow an earlier design basis. Revise the C&E settings to the 27–32 bar g band and confirm the discharge rating; the OTS model is kept as validated.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R14

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R14
Node	N2 Ethanol feed	Action party	Process
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.5	Gas back-flow into the ethanol system	P-301 stopped with XV-301 open; check valves CK301/CK302 passing; reactor at 30 bar g	H_2 and CO_2 into the pump and T-301 (design 2 bar g): overpressure of T-301 and flammable gas in its vapour space	Check valves CK301, CK302; XV-301 fails closed and closes on every trip; PSV-301 2 bar g (sized for fire only)	**8 Medium**

Recommendation

Add gas break-through from the 30 bar g reactor through failed check valves to the T-301 / PSV-301 relief cases (only fire is listed).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R15

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R15
Node	N2 Ethanol feed	Action party	Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.7	T-301 overfilled	Transfer left running; level transmitter fault	Ethanol overflow and spill (flash point ≈ 13 °C); vapour cloud near electrical equipment; fire	LAH-301 80 %; level switch LS-301 (instrument list); operator present during transfer	**9 Medium**

Recommendation

Write a T-301 filling procedure with the LS-301 high-level stop, and confirm the bund.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R16

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R16
Node	N2 Ethanol feed, N3 E-601 electric preheater, N6 D-601 HP separator, BPV-601, vent and GC, N7 D-602 LP flash and T-601 product	Action party	Safety
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.8	T-301 exposed to an external fire	Pool fire from an ethanol spill or a nearby fire	Ethanol boils off; T-301 overpressure (design 2 bar g); escalation to the skid	PSV-301 set 2 bar g, governing case external fire (543 kg/h in 2407-PE-RLB-001); fire extinguishers (assumed)	**8 Medium**
N3.4	E-601 / R-601 exposed to fire	External fire in the skid area (ethanol spill, H_2 jet fire)	Liquid in E-601 and R-601 vaporises; overpressure of E-601 and R-601 (design 36 bar g)	PSV-601 36 bar g, governing case external fire with liquid vaporisation (400.1 kg/h); no block valve between E-601 and R-601	**8 Medium**
N6.8	D-601 exposed to fire	External fire in the separator area	Gas expansion and liquid vaporisation in D-601; overpressure	PSV-603 set 36 bar g, governing case external fire (294.4 kg/h in 2407-PE-RLB-001)	**8 Medium**
N7.6	T-601 exposed to fire	Pool fire in the product area	Product boils off; T-601 overpressure	PSV-604 set 2 bar g, governing case external fire (179.8 kg/h in 2407-PE-RLB-001)	**6 Medium**

Recommendation

Cover the fire cases of T-301, R-601/E-601, D-601 and T-601 in a fire and layout review (HAZID): spacing, extinguishers, drainage.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R17

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R17
Node	N3 E-601 electric preheater	Action party	Process / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	3 Low

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N3.2	E-601 outlet too cold	Elements failing (part or all of the 6); TIC-601 failing low; TT-601 reading high	Colder liquid to the top of the bed, top-zone temperature and yield fall; no hazard	TAL-602 110 °C (re-based for this unit, assumed; C&E has TAL 143 / TALL 135 °C); JI-601 power indication	3 Low

Recommendation

Confirm the TT-602 alarm settings: the C&E TAL 143 / TALL 135 °C sit above the 127 °C design inlet; the OTS uses a re-based 110 °C (assumed).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R18

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R18
Node	N3 E-601 electric preheater	Action party	Safety / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	10 High

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N3.5	H_2 leak at a hot flange or fitting	Thermal cycling of the E-601 and R-601 connections; gasket failure; fitting not tightened after maintenance	H_2-rich gas at 30 bar g and ≈ 150 °C escapes; jet fire or explosion in the skid; flame almost invisible	Leak test before start-up (procedure, assumed); PAL-601 29 bar g and PALL-601 trip at 27 bar g (2 s delay, armed in RUN only) for a large leak only; gas detectors GD-001/GD-002 read ppm on the plant HMI (alarm to the operator; type, location and set points not in the documents); no automatic trip on gas detection (no detector row on the C&E)	**10 High**

Recommendation

Document the existing gas detectors GD-001/GD-002 (type, gas, location, alarm set points; they read ppm on the plant HMI), confirm they cover E-601, R-601, D-601 and the gas packs, decide whether a detector should trip the feeds (no detector row on the C&E today), and leak-test with N_2 (then He) before every H_2 start-up.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R19

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R19
Node	N4 R-601 trickle bed and band heaters	Action party	Instrument / Process
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	12 High

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.1	Middle zone too hot	E2 band-heater SCR failing on at full power; TIC-612 output failing high	Hot spot in the middle of the bed far above the dew point: the bed dries, the ester route stops, Cu sinters (permanent loss of activity); tube metal above the 180 °C design temperature at 30 bar g with H_2: loss of strength, possible loss of containment and H_2 fire	TAH-605 158 °C and TAHH-605 trip at 167 °C (2 s delay, opens XV-603) on the bottom TC only; TT-604 has no alarm; TIC-612 wall high alarm 260 °C (assumed); band heaters off on a trip (assumed); PSV-601 36 bar g	**12 High**
N4.2	Top zone too hot	E1 band-heater SCR or TIC-611 output failing high; TIC-611 in MAN at a high output	Hot spot in the top zone below TT-603; bed dries and sinters; wall far above the design temperature; the bottom TC and TT-603 hardly move, so nothing alarms	TIC-611 wall high alarm 260 °C (assumed); TT-603 indication (no alarm); TAHH-605 trip at 167 °C (2 s delay, opens XV-603) (bottom only)	**12 High**
N4.3	Bed hotter than indicated in cascade	TT-604 reading low while TIC-604 is in cascade on TIC-612 (thermocouple drift, junction moved); wrong cascade set point	TIC-604 raises the E2 wall to correct a false low reading; the real mid-bed goes above the dew point and the walls well above design; no alarm	TIC-604 normally in MAN; TT-603/TT-605 neighbours on the display; TAH-605 158 °C	**9 Medium**
N4.4	Zone too cold	E1 SCR failing off or a heater element open circuit; insulation damaged; TIC-611 reading high	Top of the bed cools, more liquid hold-up, yield falls; no hazard	JI-611 power indication (assumed tag); TT-603 trend; AY-620 yield	**3 Low**

Recommendation

Add high alarms on TT-603 and TT-604 and a 2oo3 high-high bed trip on TT-603/604/605 (today only TT-605 trips and TT-604 has no alarm).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R20

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R20
Node	N4 R-601 trickle bed and band heaters, N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Process / Electrical
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	12 High

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.1	Middle zone too hot	E2 band-heater SCR failing on at full power; TIC-612 output failing high	Hot spot in the middle of the bed far above the dew point: the bed dries, the ester route stops, Cu sinters (permanent loss of activity); tube metal above the 180 °C design temperature at 30 bar g with H_2: loss of strength, possible loss of containment and H_2 fire	TAH-605 158 °C and TAHH-605 trip at 167 °C (2 s delay, opens XV-603) on the bottom TC only; TT-604 has no alarm; TIC-612 wall high alarm 260 °C (assumed); band heaters off on a trip (assumed); PSV-601 36 bar g	**12 High**
N4.2	Top zone too hot	E1 band-heater SCR or TIC-611 output failing high; TIC-611 in MAN at a high output	Hot spot in the top zone below TT-603; bed dries and sinters; wall far above the design temperature; the bottom TC and TT-603 hardly move, so nothing alarms	TIC-611 wall high alarm 260 °C (assumed); TT-603 indication (no alarm); TAHH-605 trip at 167 °C (2 s delay, opens XV-603) (bottom only)	**12 High**
N9.6	Heaters left on after the feeds stop	Shutdown steps out of order; heater switched back on	Dry bed and empty E-601 heated: wall and element above design temperature	Shutdown step "Heaters off"; TAH-601; TIC-611/612/613 wall alarms (assumed); heaters trip with the SIS	**6 Medium**

Recommendation

Set the band-heater wall high alarm and trip at or below the 180 °C R-601 design temperature (now 260 °C, assumed) with an independent over-temperature cut-out per zone; review the E1 wall set point, which equals the design temperature.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R21

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R21
Node	N4 R-601 trickle bed and band heaters	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	12 High

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.2	Top zone too hot	E1 band-heater SCR or TIC-611 output failing high; TIC-611 in MAN at a high output	Hot spot in the top zone below TT-603; bed dries and sinters; wall far above the design temperature; the bottom TC and TT-603 hardly move, so nothing alarms	TIC-611 wall high alarm 260 °C (assumed); TT-603 indication (no alarm); TAHH-605 trip at 167 °C (2 s delay, opens XV-603) (bottom only)	**12 High**
N4.5	Only part of the bed wetted	Liquid channelling or a poor distributor; low ethanol rate; tilt of the tube	Dry zones run hotter and lose the ester route; yield falls; local hot spots between the thermocouples	TT-603/604/605 profile; AY-620 yield soft sensor; liquid distributor (design)	**6 Medium**

Recommendation

Fit a multipoint thermocouple in the bed so hot spots between TT-603, TT-604 and TT-605 are seen.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R22

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R22
Node	N4 R-601 trickle bed and band heaters	Action party	Control
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.3	Bed hotter than indicated in cascade	TT-604 reading low while TIC-604 is in cascade on TIC-612 (thermocouple drift, junction moved); wrong cascade set point	TIC-604 raises the E2 wall to correct a false low reading; the real mid-bed goes above the dew point and the walls well above design; no alarm	TIC-604 normally in MAN; TT-603/TT-605 neighbours on the display; TAH-605 158 °C	**9 Medium**

Recommendation

Add a deviation alarm between TT-603/604/605 and allow TIC-604 cascade only while it is healthy.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R23

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R23
Node	N5 F-601 filter, E-602 condenser and chilled water	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	6 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N5.6	Restricted flow through F-601	Filter plugging with catalyst fines or coke	Pressure drop rises: PT-601 rises against PT-608; filter element may collapse and release fines downstream	PAH-601 32 bar g; PAHH-601 trip at 33 bar g (1 s delay, opens XV-603); no differential-pressure alarm across F-601	**6 Medium**

Recommendation

Add differential-pressure indication across F-601 (PT-601 − PT-608) with a high alarm.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R24

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R24
Node	N5 F-601 filter, E-602 condenser and chilled water	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N5.3	Chilled water too warm	Chiller compressor trip with the pump still running; chiller undersized on a hot day	Condensate temperature climbs slowly; methanol and ethanol lost to the vent; no trip for a long time	TAH-609 32 °C; TAHH-609 trip at 34 °C (5 s delay, opens XV-603); no chiller-status or chilled-water supply temperature alarm	**8 Medium**

Recommendation

Add a chiller run-status alarm and a chilled-water supply temperature transmitter with a high alarm.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R25

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R25
Node	N5 F-601 filter, E-602 condenser and chilled water	Action party	Process / Mechanical
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N5.5	Process gas into the chilled-water side	E-602 tube leak or rupture (30 bar g tube side into the low-pressure shell)	H_2 and CO_2 into the chilled-water loop; shell and chiller tank overpressure; flammable gas released at the tank	PSV-602 set 4 bar g, governing case tube rupture (sized by the vendor, 2407-PE-RLB-001)	**8 Medium**

Recommendation

Confirm the E-602 shell design pressure (10 bar g in 2407-ME-EQL-001, 4 bar g in 2407-PE-RLB-001) and the PSV-602 tube-rupture sizing; vent the chilled-water tank to a safe place.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R26

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R26
Node	N6 D-601 HP separator, BPV-601, vent and GC	Action party	Safety
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.5	Flammable and toxic gas at the vent	Normal operation: H_2-rich vent gas with CO and solvent vapour; more during a depressure through XV-603 or a PSV lift	Flammable plume at the vent outlet (ignition by lightning or static: vent fire); CO exposure near the vent and the GC	Cold vent to a safe location (assumed); FT-603 indication; GC AI-612 CO	**8 Medium**

Recommendation

Carry out a vent dispersion study and hazardous-area classification for the cold vent (H_2-rich gas with CO), including depressure and PSV lifts.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R27

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R27
Node	N6 D-601 HP separator, BPV-601, vent and GC	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.3	D-601 level high	XV-601 stuck closed; LT-601 frozen at its last value; LIC-601 in MAN	Liquid carried into BPV-601, the vent line and the GC sample; ethanol liquid at the cold vent; the level rises slowly because the liquid make is small, so the alarm comes long after the cause	LAH-601 60 %; LAHH-601 trip at 80 % (5 s delay, opens XV-603); LAHH-605 trip at 85 % (5 s delay, opens XV-603) on the independent level switch LS-605; sight glass LG-601	**9 Medium**
N6.4	D-601 level low or empty	XV-601 passing or held open; LIC-601 switching levels wrong; LT-601 reading high	D-601 empties and HP gas blows through XV-601 into D-602 (design 6 bar g)	LAL-601 20 %; LALL-601 trip at 10 % (5 s delay, armed in RUN only, opens XV-603); the same XV-601 is the control and the trip valve	**9 Medium**

Recommendation

Give LIC-601 and the LALL-601 trip separate final elements (XV-601 is shared by the on/off level control and the trip).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R28

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R28
Node	N6 D-601 HP separator, BPV-601, vent and GC	Action party	Instrument / Laboratory
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	3 Low

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.6	GC reports a wrong vent composition	Wrong calibration factor for CO; sample line wet or blocked	CO and yield reported wrong; CO exposure underestimated; wrong decisions on the feed ratio	Calibration routine (assumed); AY-620 soft sensor as a cross-check	3 Low

Recommendation

Calibrate the GC with a certified CO standard (not the CO_2 factor).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R29

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R29
Node	N4 R-601 trickle bed and band heaters, N6 D-601 HP separator, BPV-601, vent and GC	Action party	Process / Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.7	Reduced catalyst exposed to air	Air ingress while the bed is warm after a shutdown or during unloading; N_2 blanket lost	Pyrophoric re-oxidation of reduced Cu: fast temperature rise, catalyst damage, ignition of residual ethanol	Shutdown procedure cools under N_2 to TT-605 < 60 °C before depressuring; XV-401 fails open	**8 Medium**
N6.7	Air drawn back from the vent	Unit depressured and cooling with the N_2 stopped (gas contracts); vent open to atmosphere	Air into D-601, E-602 and R-601: O_2 meets residual H_2 and reduced catalyst (see N4.7)	BPV-601 closes below its set pressure; N_2 blanket in the shutdown procedure (assumed)	**4 Low**

Recommendation

Keep a positive N_2 blanket after shutdown and before unloading; review the vent for back-flow of air.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R30

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R30
Node	N2 Ethanol feed, N4 R-601 trickle bed and band heaters, N5 F-601 filter, E-602 condenser and chilled water, N6 D-601 HP separator, BPV-601, vent and GC, N7 D-602 LP flash and T-601 product, N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Control / Safety
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.1	No ethanol flow	P-301 motor trip or power dip; XV-301 closing; T-301 empty (pump runs dry); suction strainer blocked	The bed dries: the wetted ester route stops and the evaporative cooling is lost; H_2 and CO_2 continue over a dry hot Cu bed; yield falls	FAL-301 at 95 % of SP; PAL-301 53 bar g; PALL-301 trip at 50 bar g (2 s delay, armed in RUN only); LAL-301 20 %; TAH-605 158 °C	**8 Medium**
N4.6	R-601 pressure low	BPV-601 drifting low (spring relaxing); BPV set wrong; leak	Dew point falls with pressure, so the bed dries; yield falls; larger vent flow	PAL-601 29 bar g; PALL-601 trip at 27 bar g (2 s delay, armed in RUN only)	**6 Medium**
N5.1	No chilled-water flow	Chilled-water pump trip; valve closed; power loss	Hot vapour reaches D-601: condensables go to the cold vent with the gas (flammable vapour release), D-601 heats up, product lost	FAL-701 63.65 L/min; FALL-701 trip at 60.3 L/min (5 s delay, armed in RUN only, opens XV-603); TAH-609 32 °C; TAHH-609 trip at 34 °C (5 s delay, opens XV-603)	**9 Medium**
N6.2	D-601 pressure low	BPV-601 spring relaxing or set low; XV-603 opened in error or on air loss (fails open); PSV passing	Depressure of the loop; the bed dries; flashing in D-601; a high gas rate to the vent	PAL-601 29 bar g; PALL-601 trip at 27 bar g (2 s delay, armed in RUN only)	**6 Medium**
N6.4	D-601 level low or empty	XV-601 passing or held open; LIC-601 switching levels wrong; LT-601 reading high	D-601 empties and HP gas blows through XV-601 into D-602 (design 6 bar g)	LAL-601 20 %; LALL-601 trip at 10 % (5 s delay, armed in RUN only, opens XV-603); the same XV-601 is the control and the trip valve	**9 Medium**
N7.1	Gas blow-by into D-602	D-601 drained with XV-601 open (passing, LIC-601 in MAN) while the low-low trips are bypassed in READY at start-up	HP gas into the 6 bar g D-602: overpressure, PSV-606 lifts; flammable gas to T-601 and the area	LALL-601 trip at 10 % (5 s delay, armed in RUN only, opens XV-603) (armed in RUN only); LAHH-602 trip at 80 % (5 s delay) (armed always); BPV-602; PSV-606 2 bar g (relief cases blow-by 22.72 kg/h and fire)	**9 Medium**
N7.3	D-602 empty	XV-602 passing or held open	Gas from D-602 to the atmospheric T-601; solvent vapour in the tank area	LAL-602 20 %; LALL-602 trip at 10 % (5 s delay, armed in RUN only)	**6 Medium**
N9.3	SIS left in READY	Arming forgotten after line-out; permissive not met and nobody follows up	The low-low trips (PALL-301, PALL-601, LALL-601, LALL-602, FALL-701) stay bypassed while the unit runs	Start-up procedure step "Arm the SIS (RUN)"; SIS state on the DCS	**9 Medium**

Recommendation

Review the start-up bypass: the low-low trips (PALL-301, PALL-601, LALL-601, LALL-602, FALL-701) are bypassed in READY; arm them automatically when their condition is healthy or after a time limit, and alarm when the SIS stays in READY with feed on.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			

Sign-off	Name	Signature	Date
HAZOP leader			
Approved (project)			

Action close-out sheet R31

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R31
Node	N7 D-602 LP flash and T-601 product	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N7.4	T-601 overfilled	P-302 not started; level indication wrong; long run without transfer	Methanol and ethanol spill (toxic and flammable) around T-601	LAH-603 80 % (C&E row 043, alarm only); level switches LS-603/604 (instrument list); PSV-604	**9 Medium**

Recommendation

Add a T-601 high-high level that closes XV-602.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R32

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R32
Node	N7 D-602 LP flash and T-601 product	Action party	Operations / Safety
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N7.5	Operator exposed to methanol	Open sampling or drumming of the product; leak at the P-302 connection	Methanol vapour and skin contact (toxic); ethanol fire risk	PPE and sampling practice (assumed)	**9 Medium**

Recommendation

Provide closed sampling for D-602/T-601 product and a methanol exposure procedure.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R33

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R33
Node	N8 Utilities: instrument air and electrical power	Action party	Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	6 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N8.1	No instrument air	Air compressor trip; header leak; PR-901 failing closed	All air-operated valves go to their fail positions: feeds and drains close, XV-401 opens the N_2 purge and XV-603 depressures; the unit trips on the first transmitter that reacts, which may not be the air header	PAL-902 1.2 bar g; PALL-902 trip at 0.9 bar g (2 s delay); fail positions: XV-101/201/301 fail closed, XV-401 and FV-401 fail open, XV-601/602 fail closed, XV-603 fails open	**6 Medium**

Recommendation

Set PALL-902 above the pressure at which the SOVs drop out (≈ 2 bar g in the model, assumed), so the air trip comes first and the first-out shows the air loss.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R34

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R34
Node	N8 Utilities: instrument air and electrical power	Action party	Electrical / Instrument
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N8.3	No electrical power	Site supply failure; main breaker trip	Heaters, P-301, chiller and chilled-water pump stop together; the SIS de-energises to trip; if the SIS, MFC or GC power is lost there is no indication of the state of the unit	De-energise-to-trip SIS (assumed); fail-safe valves (fail positions: XV-101/201/301 fail closed, XV-401 and FV-401 fail open, XV-601/602 fail closed, XV-603 fails open); N_2 purge from the packs needs no power	**9 Medium**
N8.4	Voltage dip	Grid disturbance	Motors (P-301, chilled-water pump) drop out while the heaters ride through: loss of ethanol or cooling with the heat on	PALL-301 trip at 50 bar g (2 s delay, armed in RUN only); FALL-701 trip at 60.3 L/min (5 s delay, armed in RUN only, opens XV-603); TAH-605 158 °C	**6 Medium**

Recommendation

Confirm UPS supply for the SIS, DCS, MFCs and GC, and document the behaviour of every final element on loss of power.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R35

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R35
Node	N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	10 High

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.1	H_2 admitted before the air is purged out	Start-up after maintenance with the unit only pressurised with N_2, not purged to a low O_2 content; step skipped	H_2/air flammable mixture in R-601 over hot catalyst: explosion or fast exotherm	Start-up procedure pressurises with N_2 to PT-601 ≥ 29 bar g before heating; no O_2 measurement	**10 High**

Recommendation

Write a pressure-swing N_2 purge procedure with a portable O_2 check before H_2 is admitted.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R36

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R36
Node	N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Control
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	9 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.2	H_2 in before the ethanol flows	Steps done out of order; P-301 not started or not delivering	H_2 and CO_2 over a dry hot Cu bed: no evaporative cooling, hot spots, catalyst damage	Start-up step order (ethanol, then H_2, then CO_2); TAH-605 158 °C; PALL-301 bypassed in READY	**9 Medium**

Recommendation

Add a permissive: ethanol flowing (FT-301 above its low limit) before XV-201 can open in READY.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R37

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R37
Node	N1 H_2, CO_2 and N_2 gas feed, N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.8	Wrong gas connected to a feed line	Wrong pack or cylinder connected at changeover (air, O_2 or another gas on the H_2 or CO_2 line); pigtail not purged	Air or O_2 with H_2 over hot reduced Cu catalyst: flammable mixture, fast exotherm, catalyst oxidation, possible explosion in R-601	Gas-specific pack connections (assumed); pack labels; GC on the vent gas (after the event)	**5 Medium**
N9.8	Air in the H_2 manifold after a changeover	Pigtail not vent-purged after connecting the new pack	Air slug with H_2 to the reactor (see N1.8)	Pack-change practice (assumed)	**8 Medium**

Recommendation

Vent-purge the pigtail with N_2 at every pack changeover (written step).

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R38

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R38
Node	N4 R-601 trickle bed and band heaters, N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Process / Operations
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.7	Reduced catalyst exposed to air	Air ingress while the bed is warm after a shutdown or during unloading; N_2 blanket lost	Pyrophoric re-oxidation of reduced Cu: fast temperature rise, catalyst damage, ignition of residual ethanol	Shutdown procedure cools under N_2 to TT-605 < 60 °C before depressuring; XV-401 fails open	**8 Medium**
N9.9	Catalyst loading, reduction or unloading done wrongly	No written procedure in the unit documents; reduction with too much H_2 or too fast; unloading while the catalyst is reduced and warm	Reduction exotherm sinters the catalyst; pyrophoric catalyst on unloading; dust exposure	None documented	**8 Medium**

Recommendation

Write and HAZOP the catalyst loading, reduction and passivation/unloading procedures before first use.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R39

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R39
Node	N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Operations / Safety
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	8 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.10	Equipment opened before it is gas-free	Isolation and purge incomplete; residual H_2 or ethanol in a dead leg	Flammable or toxic release at the opened joint; fire	Permit to work (assumed)	**8 Medium**

Recommendation

Use a permit to work with N_2 purge and a gas test before any equipment is opened.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			

Action close-out sheet R40

HAZOP phase	Project HAZOP (teaching study), Rev A	Action no.	R40
Node	N9 Procedures: purge, start-up, shutdown, pack changeover, catalyst	Action party	Operations / Training
Drawings	P&ID 2407-PE-PID-001 A6; C&E 2407-IN-CE-001 A2	Highest risk	6 Medium

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.11	Emergency shutdown late	Operator hesitates; ESD-SW not reachable	An escalating event (leak, fire) is fed longer	ESD-SW (C&E row 001) with the full trip action: closes XV-101/201/301 and FV-101/201, stops P-301, opens XV-401 and FV-401 (N_2 purge), E-601 and the band heaters off; XV-603 opens	**6 Medium**

Recommendation

Train operators on the simulator on the deviations of section 9 and on the start-up bypass, and record their response times.

Response of the action party

Status	☐ Open ☐ Accepted as proposed ☐ Accepted with alternative ☐ Rejected (reason given) ☐ Closed

Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Approved (project)			